

Magnesium modulates phospholipid metabolism to promote bacterial phenotypic resistance to antibiotics

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Hui Li, Jun Yang, Su-fang Kuang, Bo Peng 

State Key Laboratory of Bio-Control, School of Life Sciences, Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai), Guangdong Province Key Laboratory for Pharmaceutical Functional Genes, Sun Yat-sen University, Higher Education Mega Center, Guangzhou 510006, People's Republic of China • Laboratory for Marine Biology and Biotechnology, Marine Fisheries Science and Food Production Processes, Qingdao National Laboratory for Marine Science and Technology, Qingdao 266071, China

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Abstract

Non-inheritable antibiotic resistance or phenotypic resistance ensures bacterial survival upon antibiotic treatment. However, exogenous factors in promoting phenotypic resistance is poorly defined. Here, we demonstrate that *Vibrio alginolyticus* are recalcitrant to killing by a broad spectrum of antibiotics under high magnesium. Functional metabolomics demonstrate that magnetism modulates the biosynthesis of fatty acids in increasing the biosynthesis of saturated fatty acids while decreasing unsaturated fatty acids. Exogenous supplementation of fatty acids confirm the role of fatty acids in antibiotic resistance. Furthermore, functional lipidomics reveal that glycerophospholipid metabolism is the major metabolic pathway remodeled by magnetism, where the biosynthesis of PE is decreased but PG is increased. Thus, the membrane composition is altered, leading to increased membrane polarization, and decreased permeability and fluidity. These together reduce the uptake of antibiotics by the bacteria. Thus, our study suggest a previously unrecognized metabolic mechanism by which bacteria escape antibiotic killing by utilizing environmental factor.

eLife assessment

The study explored the influence of magnesium on phenotypic antibiotic resistance in two *Vibrio* model bacteria. This research is **fundamental** for revealing the phenotypic antibiotic resistance mechanism utilized by the specified model bacteria in elevated levels of magnesium. The study produced **solid** evidence indicating that in high concentrations of magnesium, the efficacy of selected antibiotics was diminished due to decreased biosynthesis of unsaturated fatty acids and PE, along with an increase in the biosynthesis of PG.

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Introduction

Non-inheritable antibiotic resistance or phenotypic resistance represent a serious therapeutic challenge for bacterial infectious diseases. Phenotypic resistance does not involve genetic mutations, instead, this type of resistance are transient. Thus, bacteria resume to normal growth after removing the antibiotics. Two extensively studied phenotypic resistance includes biofilm and bacterial persisters (Corona & Martinez, 2013 [\[1\]](#)). Biofilms have complex structures that impede the diffusion of antibiotics and contain elements sequestering antibiotics to prevent their further actions (Ciofu *et al*, 2022 [\[2\]](#)). Moreover, biofilm-forming bacteria and persisters themselves have distinct metabolic state that also significantly negatively influence their susceptibility to antibiotics (Yan & Bassler, 2019 [\[3\]](#)). These two types of phenotypic resistance have common phenomena that their growth are retarded or even stopped in the presence of antibiotics (Corona & Martinez, 2013 [\[1\]](#)). However, factors promoting bacterial phenotypic resistance that is still proliferate in the presence of antibiotics are poorly defined.

There is a growing body of evidence that metal ions have diverse impact on chemical, physical and physiological phenomena that play roles in development of antibiotic resistance (Booth *et al*, 2011 [\[4\]](#); Lu *et al*, 2020 [\[5\]](#); Poole, 2017 [\[6\]](#)). One mechanism could involve genetic elements that harbor genes conferring resistance to metals and antibiotics (Poole, 2017 [\[6\]](#)). On the other hand, recent reports suggest that metal cations directly hinder (or enhance) the activity of specific antibiotic drugs (Zhang *et al*, 2014 [\[7\]](#)) and that the metabolic environment impacts bacterial sensitivity to antibiotics (Jiang *et al*, 2023 [\[8\]](#); Lee & Collins, 2012 [\[9\]](#); Peng *et al*, 2015 [\[10\]](#); Zhang *et al*, 2020 [\[11\]](#); Zhao *et al*, 2021 [\[12\]](#)). Light metal ions, such as magnesium, sodium, and potassium, serve as cofactors to many enzymes (Du *et al*, 2016 [\[13\]](#)), some of which could influence drug efficacy. Heavy metal ions, including Cu^{2+} and Zn^{2+} , confer resistance to antibiotics (Yazdankhah *et al*, 2014 [\[14\]](#); Zhang *et al*, 2018 [\[15\]](#)). Recent reports suggest that sodium negatively regulates redox states to promote antibiotic resistance in *Vibrio alginolyticus* (Yang *et al*, 2018 [\[16\]](#)), while actively growing *Bacillus subtilis* cope with ribosome-targeting antibiotics based on ion flux modulation (Lee *et al*, 2019 [\[17\]](#)). But whether such ions can promote phenotypic resistance is less explored.

Results

Mg²⁺ promotes resistance to antibiotics

Magnesium is rich in marine and marine aquaculture, where antibiotics are commonly used. We directly investigate the effect of Mg^{2+} on balofloxacin (BLFX) susceptibility in two strains of *Vibrios*, *V. alginolyticus* ATCC33787 and *V. parahaemolyticus* VP01, which were isolated from marine aquaculture. In the initial experiment, the ionic environment of marine water was simulated by adding 39 mM NaCl, 35 mM Mg_2SO_4 , 7 mM KCl, and 7 mM CaCl_2 to LB medium (Table EV1), to create a medium which we designated “artificial seawater” (ASWT). The minimum inhibitory concentrations (MIC) for BLFX in ATCC33787 and VP01 were 54 $\mu\text{g/mL}$ and 25 $\mu\text{g/mL}$ in ASWT and 1 $\mu\text{g/mL}$ and 3 $\mu\text{g/mL}$ in LB medium, respectively (Fig 1A [\[18\]](#)). The role of exogenous NaCl in antibiotic resistance was investigated in a previous study (Yang *et al*, 2018 [\[16\]](#)); and the MIC for BLFX was the same in LB medium or M9 medium plus 7 mM KCl or 7 mM CaCl_2 (Fig EV1 [\[19\]](#)). However, the MIC for BLFX was higher in M9 medium plus Mg_2SO_4 or MgCl_2 (Fig 1B [\[20\]](#)) and the MIC for BLFX increased with increasing concentration of MgCl_2 . Specifically, addition of 50 mM or 200 mM MgCl_2 increased the MIC for BLFX 200- or 1600-fold, respectively (Fig 1B [\[20\]](#)). Using the Oxford cup method, the zone of inhibition increased with increasing MgCl_2 from 15.6 mM – 62.4 mM (Fig 1D [\[21\]](#)). Exogenous MgCl_2 also increased MICs for other quinolones (*e.g.* nalidixic acid, levofloxacin, ciprofloxacin, ofloxacin, and moxifloxacin (Fig 1E [\[22\]](#)) and non-quinolone antibiotics

including antibacterial peptides (colistin), macrolides (roxithromycin), tetracyclines (oxytetracycline), β -lactams (ceftriaxone, ceftazidime), and aminoglycosides (amikacin, kanamycin, and gentamicin) (**Fig 1F**). Such non-specific resistance promoting effects indicate that Mg^{2+} promotes phenotypic resistance.

Extracellular Mg^{2+} , intracellular Mg^{2+} and chelation of antibiotic

Previous reports suggest that chelation of antibiotics by magnesium ions inhibits antibiotic uptake (Deitchman *et al*, 2018; Lunestad & Goksøyr, 1990). To explore this possibility, the zone of inhibition for balofloxacin was measured with various concentrations of $MgCl_2$, ranging from 0 mM to 200 mM. Marine water usually contains 50 mM $MgCl_2$. Lower concentrations of $MgCl_2$ (0.78, 3.125 or 12.5 mM) did not alter the zone of inhibition, while higher concentration including 50 and 200 mM $MgCl_2$ decreased the zone of inhibition (**Fig 2A**). Correlation of the $MgCl_2$ concentration-dependent resistance suggests that chelation influences bacterial survival by 25.5% and 27.7% in 50 and 200 mM $MgCl_2$, respectively, but not at lower $MgCl_2$ concentrations (**Fig 2B & 2C**). Furthermore, intracellular balofloxacin decreased with increasing concentration of $MgCl_2$ (**Fig 2D**), while exogenous Mg^{2+} increased intracellular Mg^{2+} level in a dose-dependent manner, where 50 and 200 mM $MgCl_2$ increased 1.21 and 1.31 mmol/L, respectively (**Fig 2E**). We also examined rate of efflux and expression of LPS, which contribute to antibiotic resistance (Kobylka *et al*, 2020; Song *et al*, 2020). The results show that expression of *TolC/tolC* and LPS was unaffected by Mg^{2+} (**Fig 2F** and **2G**). Loss of *waaF*, *lpxA*, and *lpxC*, three key genes involved in LPS biosynthesis, did not influence sensitivity/resistance to balofloxacin (**Fig 2H**). These data demonstrate that influx of Mg^{2+} plays roles in balofloxacin resistance.

$MgCl_2$ affects bacterial metabolism

Although it is well established that Mg^{2+} regulates the activity of many metabolic enzymes and strongly influences bacterial metabolism (Garfinkel & Garfinkel, 1985), the mechanism of Mg^{2+} in promoting antibiotic resistance is less explored. Thus, *V. alginolyticus* was cultured in M9 medium in the presence of various amounts of $MgCl_2$ (0 mM, 0.78 mM, 3.125 mM, 12.5 mM, 50 mM, or 200 mM), analyzed by GC-MS, and identified 54 metabolites. Five biological and two technical replicates were included for each treatment (**Fig EV2**). Rank-sum test and a permutation test were used to identify differentially expressed metabolites. Forty-one metabolites were identified ($p < 0.05$) (**Fig 3A**), which were presented as a heatmap and Z-value in **Fig EV3**. Orthogonal partial least square discriminant analysis (OPLS-DA) was conducted that separated the six treatments into three groups: Group 1 included 0 mM, 0.78 mM, and 3.125 mM $MgCl_2$; Group 2 was a singlet of 12.5 mM; Group 3 included 50 mM and 200 mM $MgCl_2$. Component t[1] distinguishes Group 1 from Groups 2 and 3; component t[2] distinguishes Group 3 from Groups 1 and 2 (**Fig 3B**). Discriminating variables are shown as S-plot, where cut-off values are ≥ 0.05 absolute value of covariance p and 0.5 for correlation p (corr). Six biomarkers/metabolites were selected from component t[1] and p[1]. More specifically, the abundance of cadaverine, urea, palmitic acid, aminoethanol, and fumaric acid were increased, but pyroglutamic acid and glutamic acid were decreased (**Fig 3C**; **Fig EV4**). Pathway enrichment analysis suggests that twelve pathways are involved. Notably, one of the pathways is biosynthesis of unsaturated fatty acids (terminology in the software. Actually, it is biosynthesis of fatty acids) (**Fig 3D**). Interestingly, the abundance of palmitic acid and stearic acid were increased in an Mg^{2+} dose-dependent manner (**Fig 3E**; **Fig EV5**). Moreover, palmitic acid was a crucial biomarker (**Fig 3C**). Exogenous palmitic acid increased bacterial resistance to balofloxacin (**Fig 3F**). These results suggest that fatty acid metabolism may be critical to the resistance.

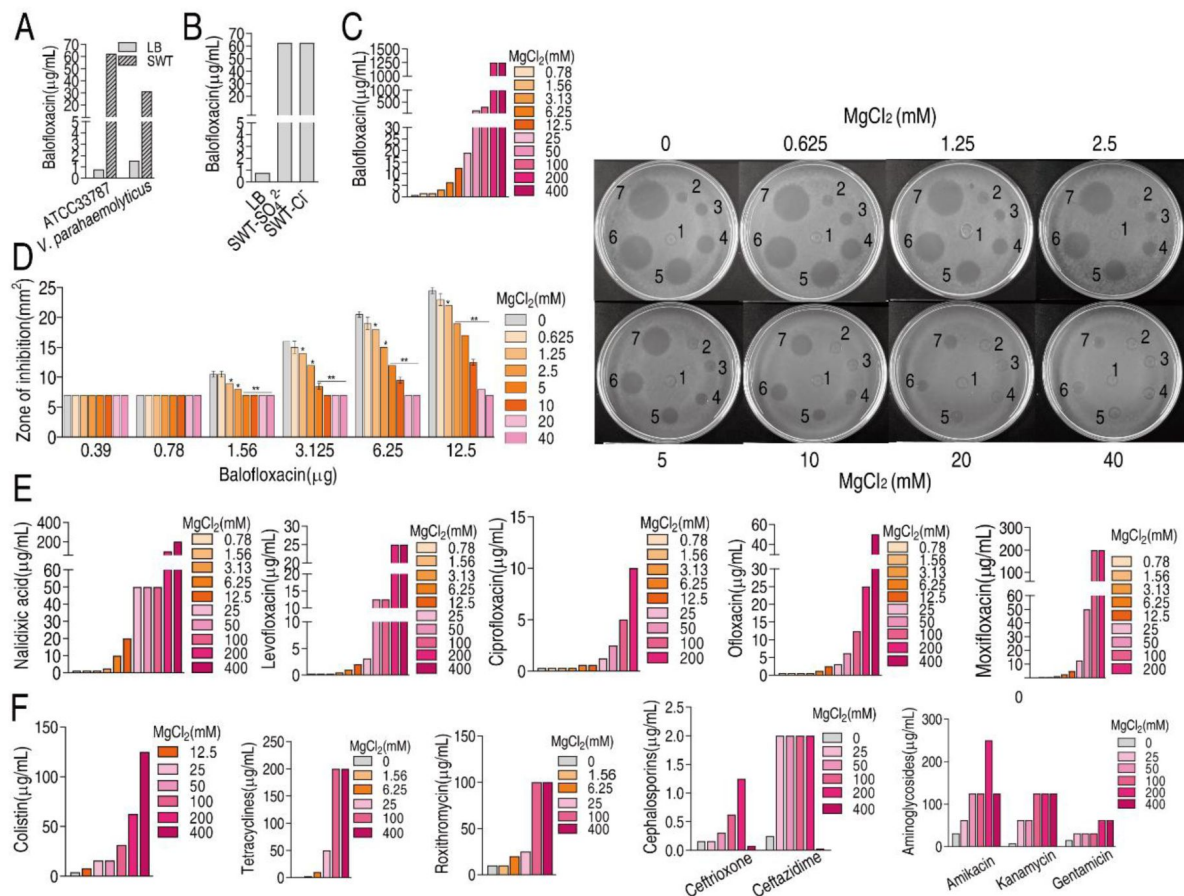


Figure 1.

Magnesium promotes bacterial resistance to antibiotics.

A. MIC of ATCC 33787 and *V. parahaemolyticus* to BLFX in SWT or LB medium by microtitre-dilution- method. **B.** MIC of ATCC 33787 to BLFX in SWT with the replacement of MgCl₂ with MgSO₄ MgCl₂ by microtitre-dilution-method. **C.** MIC of ATCC 33787 to BLFX in SWT with the indicated concentrations of MgCl₂ by microtitre-dilution-method. **D.** MIC of ATCC 33787 to BLFX in the indicated concentrations of BLFX and MgCl₂ by Oxford cup test. **E.** MIC of ATCC 33787 to other quinolones in SWT with the indicated concentrations of MgCl₂ by microtitre-dilution-method. **F.** MIC of ATCC 33787 to other classes of antibiotics in SWT with the indicated concentrations of MgCl₂ by microtitre-dilution-method.

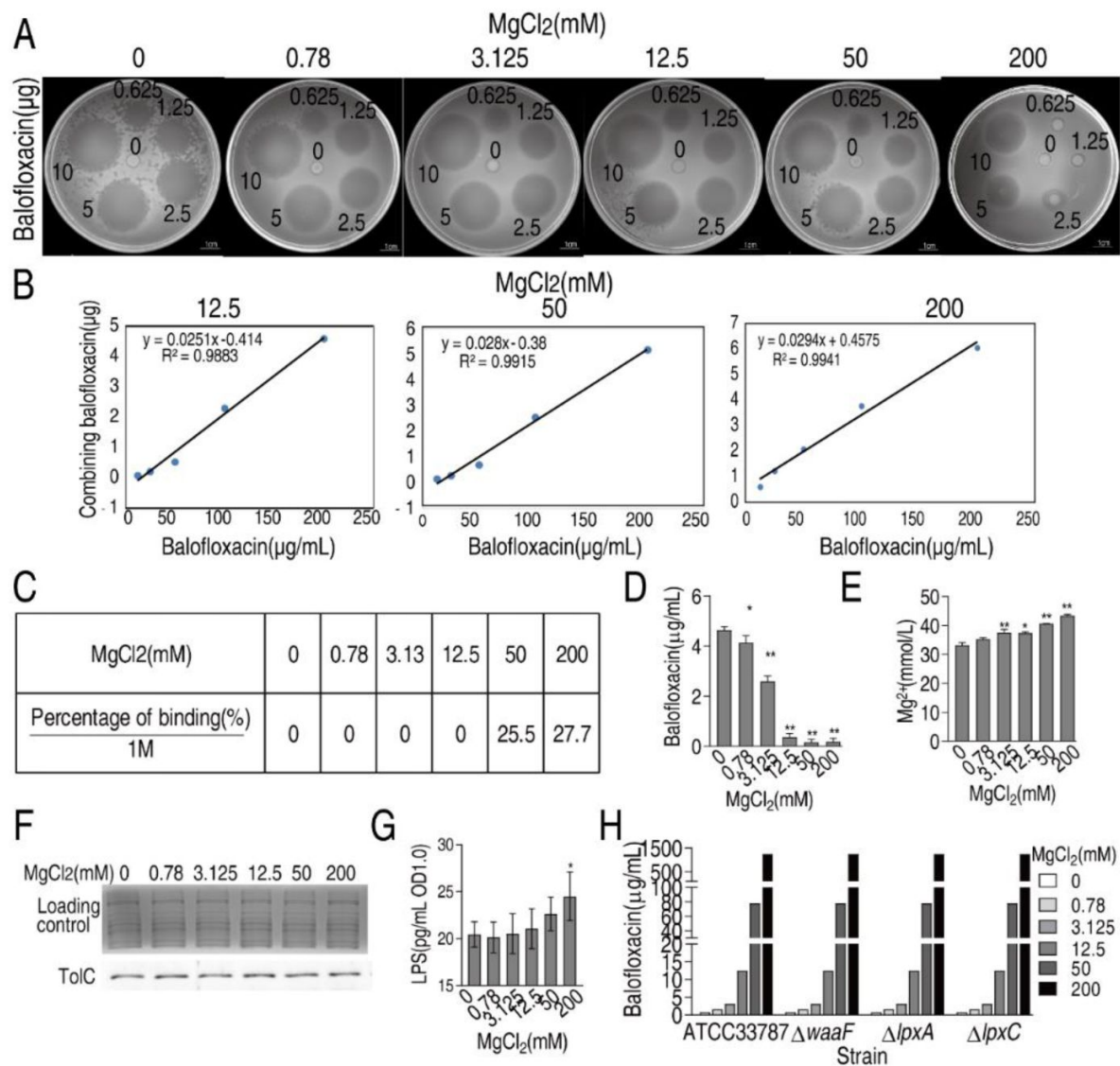


Figure 2.

Mg²⁺ elevates intracellular Mg²⁺ and promotes balofloxacin uptake.

A. Oxford cup test of 0, 12.5, 25, 50, 100, 200 µg/mL BLFX to ATCC33787 following by separately incubation with the indicated concentrations of MgCl₂ for 5 h. **B.** Inhibitory curve of data (**A**). **C.** Binding efficacy of (**A**). **D.** Intracellular BLFX of ATCC 33787 in SWT with the indicated concentrations of MgCl₂ and 60 µg/mL BLFX. **E.** Intracellular Mg²⁺ of ATCC 33787 in SWT with the indicated concentrations of MgCl₂. **F.** Western blot for abundance of TolC in the presence of MgCl₂. **G.** LPS quantification at indicated concentrations of MgCl₂. **H.** MIC of ATCC 33787 and its mutants $\Delta waaF$ $\Delta lpxA$ $\Delta lpxC$ in SWT with the indicated concentrations of MgCl₂, which is measured by microtitre-dilution-method.

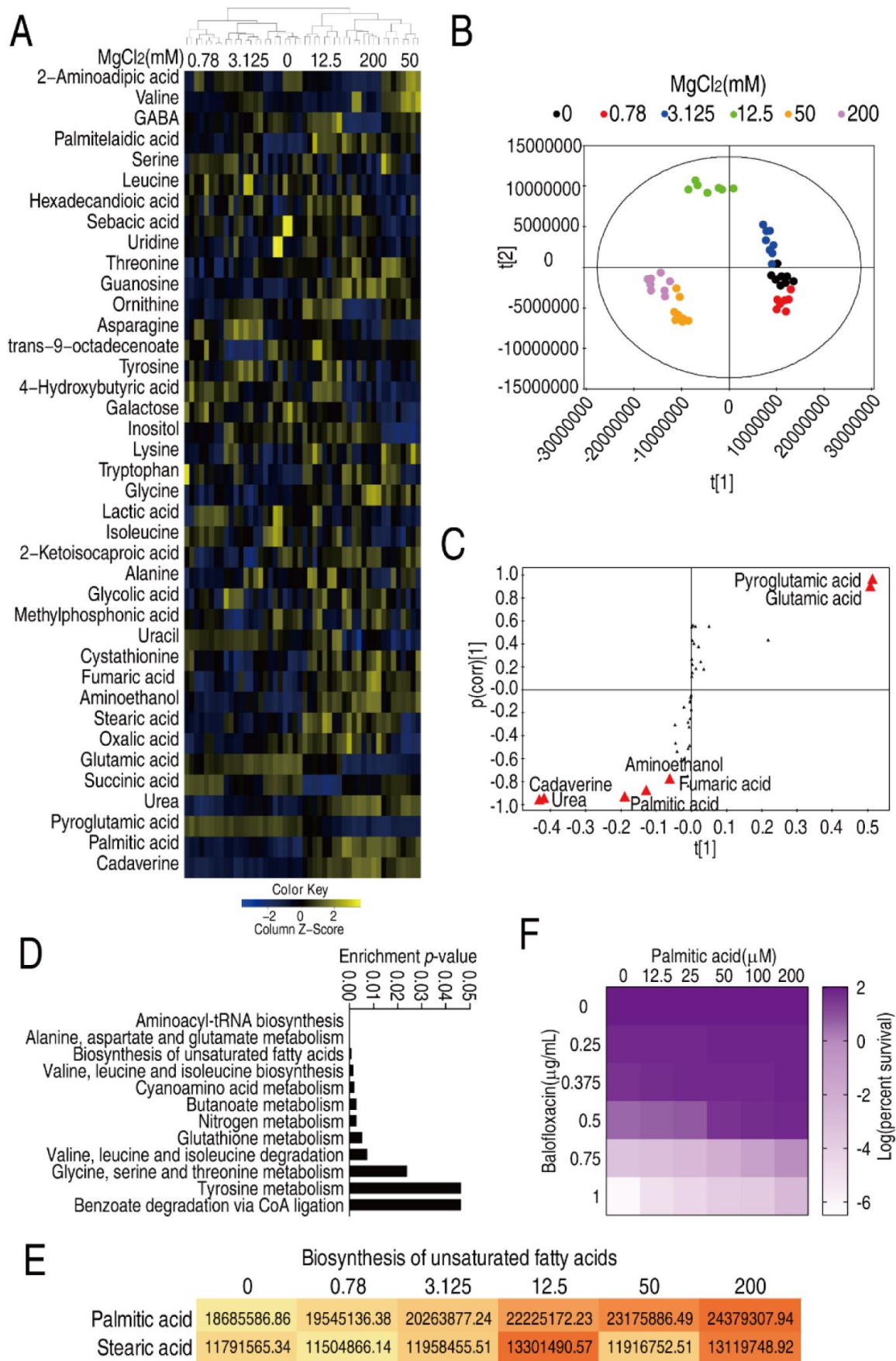


Figure 3.

Mg²⁺-induced metabolomics.

A. Differential metabolomes in the absence or presence of the indicated concentrations of MgCl₂. Yellow color and blue color indicate increase and decrease of metabolites relative to the median metabolite level, respectively (see color scale). **B.** PCA analysis of different concentrations of MgCl₂-induced metabolomes. Each dot represents the technical replicate of samples in the plot. **C.** S-plot generated from OPLS-DA. Predictive component p [1] and correlation p(corr) [1] differentiate 0, 0.78, 3.125 mM MgCl₂ from 12.5, 50, 200 mM MgCl₂. Predictive component p[2] and correlation p(corr)[2] separate 0, 0.78, 50, 200 mM MgCl₂ from 3.125, 12.5 mM MgCl₂. The triangle represents metabolites in which candidate biomarkers are marked. **D.** Enriched pathways by differential abundances of metabolites. **E.** Scatter plot of palmitic acid and stearic acid in the indicated concentrations of MgCl₂, which comes from data (**A**). **F.** Synergy analysis for alofloxacin with palmitic acid for *V. alginolyticus*. Synergy is represented using a color scale or an isobologram, which compares the dose needed for 50% inhibition for synergistic agents (white) and non-synergistic (i.e., additive) agents (purple).

Mg²⁺ regulates fatty acid biosynthesis

Fatty acids include saturated and unsaturated fatty acids whose biosynthesis shared common biosynthetic pathways from acetyl-CoA to enoyl-ACP, sequentially mediated by *fabD* or *fabH*, *fabB/F*, *fabG* and *fabA/Z*. Enoyl-ACP is the direct source to produce unsaturated fatty acids or is metabolized to Acyl-ACP via *fabV* to produce saturated fatty acids. Unsaturated fatty acids produce saturated fatty acids via *tesA/B* and *yciF*. qRT-PCR was used to quantify the expression of involved genes in bacteria treated with different concentration of MgCl₂. The expression of 18 genes was increased when treated with 50 or 200 mM MgCl₂ (**Fig 4A**). The increased expression of FabF was confirmed by Western blot (**Fig 4B**). Expression of *tesA*, *tesB*, and *yciA*, which convert unsaturated to saturated fatty acids, was increased (**Fig 4C**). The expression of *fabA* that promotes enoyl-ACP and biosynthesis of unsaturated fatty acids was decreased (**Fig 4D**), which was confirmed at protein level with Western blot (**Fig 4B**). Oxazole-2-amine and triclosan inhibit synthesis of fatty acids and decrease bacterial survival in the presence of balofloxacin (**Fig 4E**). These results indicate that Mg²⁺ inhibits synthesis of unsaturated fatty acids and promotes synthesis of saturated fatty acids.

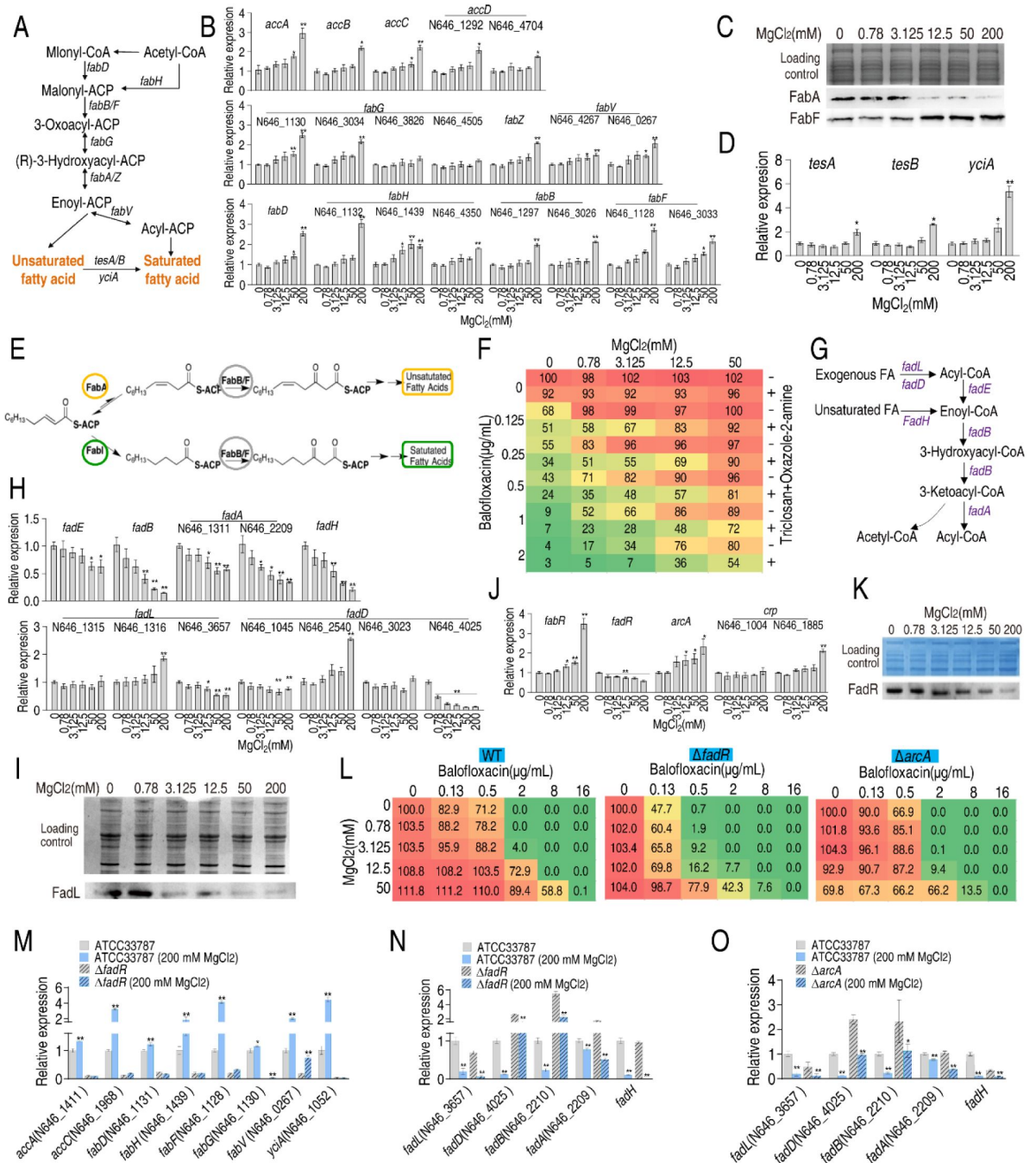


Figure 4

Mg²⁺ promotes biosynthesis of fatty acids.

A. qRT-PCR for expression of fatty acid biosynthesis genes in the absence or presence of MgCl₂. **B.** Western blot for abundance of proteins responsible for conversion to unsaturated fatty acid biosynthesis or saturated fatty acid in the absence or presence of MgCl₂. **C.** qRT-PCR for expression of genes encoding conversion from unsaturated fatty acid biosynthesis to saturated fatty acid in the absence or presence of MgCl₂. **D.** Diagram for biosynthesis of saturated and unsaturated acids. **E.** Synergy analysis for balofloxacin with triclosan + oxazole-2- amine for ATCC33787. Synergy is represented using a color scale or an isobologram, which compares the dose needed for 50% inhibition for synergistic agents (blue) and non-synergistic (i.e., additive) agents (red). **F.** Diagram for degradation of fatty acids. **G.** qRT-PCR for expression of genes encoding fatty acid degradation in the absence or presence of the indicated concentrations of MgCl₂. **H.** Western blot for abundance of FadL in the absence or presence of the indicated concentrations of MgCl₂. **I.** qRT-PCR for expression of regulatory genes of fatty acid biosynthesis in the absence or presence of the indicated concentrations of MgCl₂. **J.** Western blot for abundance of FadR in the absence or presence of the indicated concentrations of MgCl₂. **K and L.** Synergy analysis for MgCl₂ with BLFX for Δ *fadR* (L) and Δ *arcA* (M). Synergy is represented using a color scale or an isobologram, which compares the dose needed for 50% inhibition for synergistic agents (white) and non-synergistic (i.e., additive) agents (red). **M, N and O** qRT-PCR for expression of genes encoding biosynthesis of fatty acids and degradation of fatty acids (**M**) in ATCC33787 and Δ *fadR* (**N**) and Δ *arcA* (**O**) in the presence or absence of 200 mM MgCl₂.

We also quantified the gene expression of fatty acid degradation (**Fig 4F**). Interestingly, the expression of genes involved in unsaturated fatty acid degradation was decreased in an MgCl₂ dose-dependent manner (**Fig 4G**), while the expression of genes for exogenous fatty acid degradation was increased. Western blot data confirmed that the abundance of FadL decreased with increasing MgCl₂ (**Fig 4H**). FadH degrades unsaturated fatty acids, and influences the ratio of unsaturated to saturated fatty acids. In general, these results suggest that MgCl₂ inhibits degradation of fatty acids.

FabR, FadR, ArcA, and cAMP/CRP are transcriptional factors in regulating fatty acid biosynthesis. FabR inhibits biosynthesis of unsaturated fatty acids; FadR promotes biosynthesis and inhibits fatty acids degradation; ArcA and cAMP/CRP inhibit and promote fatty acid degradation, respectively (Feng & Cronan, 2010; Fujita *et al.*, 2007). The expression of *fabR* and *fadR* was increased, *arcA* was decreased. The expression of N646_1004 did not change with increasing MgCl₂, while expression of N646_1885 increased only in the presence of 200 mM MgCl₂ (**Fig 4I**). Western blot data confirmed the decreased expression of FadR with increasing MgCl₂ concentration (**Fig 4J**). Viability of Δ *fadR* and Δ *arcA* was decreased in the presence of MgCl₂, balofloxacin or both in a dose-dependent manner (**Fig 4K**). Thus, FadR and ArcA could play roles in bacterial resistance to balofloxacin, presumably by a mechanism that is linked to the abundance of saturated/unsaturated fatty acids.

Furthermore, to investigate whether Mg²⁺ functions through FadR and ArcA, expression of 8 genes of fatty acid biosynthetic genes (*accA*, *accC*, *fabD*, *fabH*, *fabR*, *fabG*, *fabV*, and *yciA*) and 5 genes of fatty acid degradation (*fadL*, *fadD*, *fadB*, and *fadA*) were quantified in wildtype, Δ *fadR* and Δ *arcA* bacterial strains. As expected, the expression of fatty acid biosynthetic genes was decreased in Δ *fadR* cells (**Fig 4L**). However, higher expression of *fadD*, *fadB*, and *fadA* (elevated *fadA* only for Δ *fadR*) was detected in Δ *fadR* or Δ *arcA* strains, but the expression of these genes was negatively regulated by 200 mM MgCl₂ (**Fig 4M-O**). These results indicate that Mg²⁺ promotes biosynthesis of fatty acids by positively regulating *fadR*, and inhibits degradation of fatty acids through *fadR*, *arcA* and their downstream targets.

MgCl₂ affects 16-carbon and 18-carbon fatty acid metabolism

To investigate the effect of Mg²⁺ on biosynthesis of saturated and unsaturated fatty acids, LC-MS was used to quantify 16-carbon and 18-carbon fatty acids, the main precursors for lipid biosynthesis (Zhang & Rock, 2008b [↗](#)). The abundance of saturated fatty acids, palmitic acid (C16:0), was increased, while the abundance of five unsaturated fatty acids, palmitoleic acid (C16:1), linoelaidic acid (C18:2), linoleic acid (C18:2), alpha-linoleic acid (C18:3), stearidonic acid (C18:4) was decreased with increasing MgCl₂ concentration (**Fig 5A** [↗](#)). The abundance of stearic acid (C16:0) and vaccenic acid (C18:1) was unaffected. Total saturated fatty acids increased and total unsaturated fatty acids decreased with increasing Mg²⁺ (**Fig 5B** [↗](#)). These results support the conclusion that Mg²⁺ upregulates biosynthesis of saturated fatty acids but downregulates biosynthesis of unsaturated fatty acids.

Furthermore, direct exposure to palmitic acid inhibited balofloxacin-mediated killing (**Fig 3F** [↗](#)) but linolenic acid promoted balofloxacin-mediated killing in a dose-dependent manner (**Fig 5C** [↗](#)). Increasing exogenous palmitic acid and linolenic acid increased intracellular palmitic acid and linolenic acid, respectively (**Fig 5D** [↗](#)). When cells were co-treated with palmitic acid, linolenic acid, and balofloxacin, the effects of the two fatty acids appeared to be antagonistic as demonstrated by Bliss model (**Fig 5E** [↗](#) & **5F** [↗](#)). These results indicate that saturated and unsaturated fatty acids have antagonistic effects on balofloxacin resistance.

Mg²⁺ promotes phospholipid biosynthesis

Fatty acids are biosynthetic precursors of lysophosphatidic acid (LPA) and phosphatidic acid (PA), two key components of the cell membrane (Zhang & Rock, 2008a [↗](#)). Thus, LC-MS was used to profile the effect of Mg²⁺ on membrane lipid composition (**Fig EV6A** [↗](#)). Increasing concentration of Mg²⁺ increased the percentage of lipids increased from 61% to 67%, and saturated fatty acids from 24% to 26%, but decreased the percentage of unsaturated fatty acids from 15% to 8% (**Fig EV6B** [↗](#)). The abundance of 11 lipids, 32 lipids, and 53 were increased and of 26 lipids, 52 lipids, and 107 lipids were decreased in 3.125 mM, 50 mM, and 200 mM MgCl₂, respectively (**Fig 6C** [↗](#)). Saturated fatty acid and lipid increased and unsaturated fatty acid decreased in an Mg²⁺ dose-dependent manner (**Fig EV6D** [↗](#)). A total of 52 lipids were quantified, including 17 high abundance lipids (**Fig EV6E** [↗](#)). Phosphatidylethanolamine (PE) and phosphatidylglycerol (PG) were the first and second highest lipids in abundance (**Fig EV6E** [↗](#)) and PE decreased while PG increased with increasing Mg²⁺ (**Figs 6A** [↗](#), **6B** [↗](#)). Similar effect was observed for unsaturated derivatives of PE and PG (**Fig 6C** [↗](#)). Principal component analysis (PCA) showed that component t [1] differentiates 0 and 3.125 mM Mg²⁺ from 50 and 200 mM Mg²⁺, while component t [2] separates 50 mM Mg²⁺ from 0 and 200 mM Mg²⁺ and variants in 3.125 mM of Mg²⁺ treatment (**Fig 6D** [↗](#)).

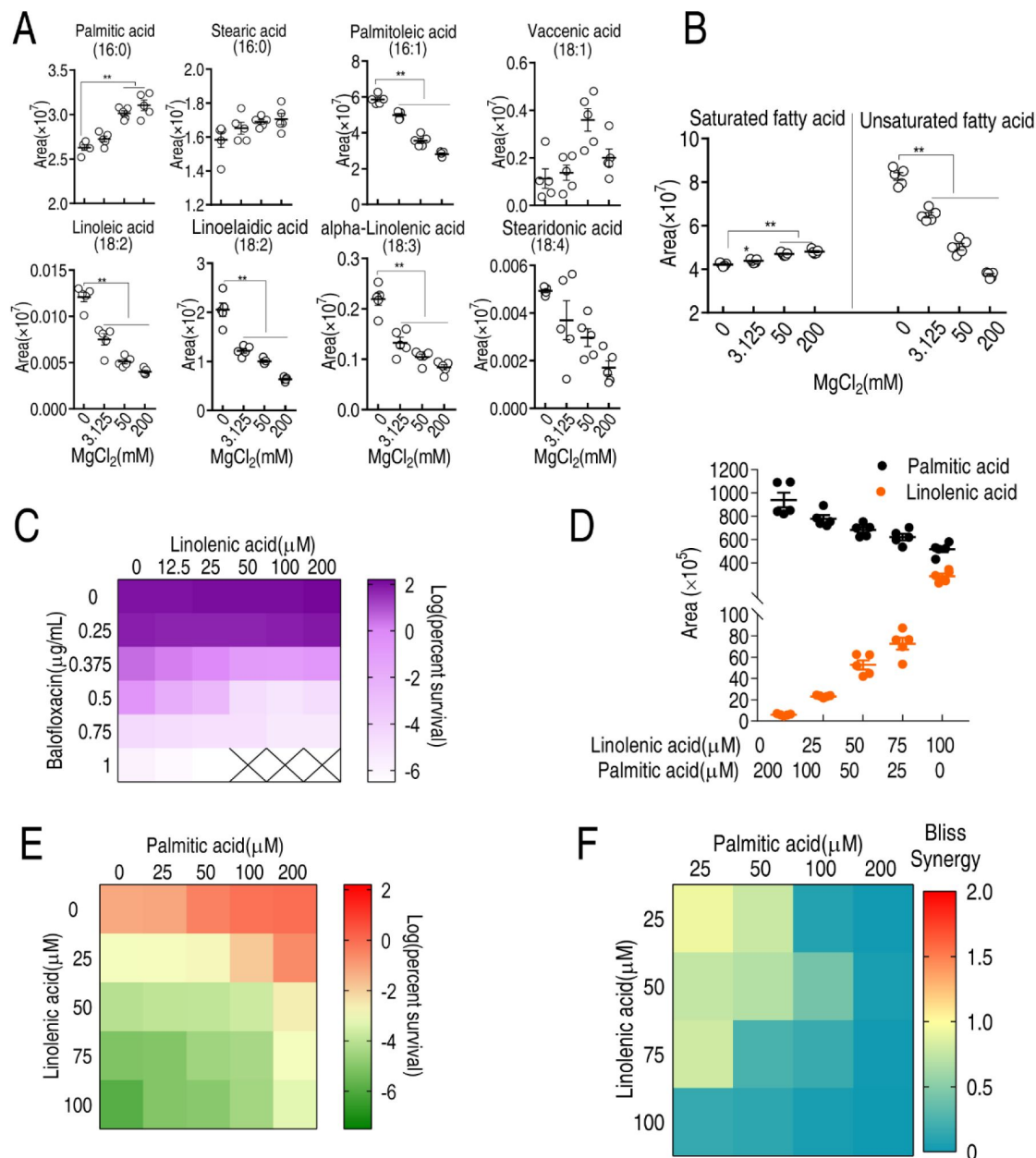


Figure 5.

LC-MS targeted 16-carbon and 18-carbon fatty acids and role of palmitic acids and linolenic acids in BLFX resistance.

A. Scatter plots of 16-carbon and 18-carbon fatty acids, detected by LC-MS. **B.** Scatter plots for total saturated fatty acid and unsaturated fatty acid with 16-carbon and 18-carbon in data (A). **C.** Synergy analysis for BLFX with linolenic acid for ATCC 33787. Synergy is represented using a color scale or an isobologram, which compares the dose needed for 50% inhibition for synergistic agents (while) and non-synergistic (i.e., additive) agents (purple). **D.** LC-MS for abundance of intracellular linolenic acid and palmitic acid of ATCC 33787 in synergy of the indicated exogenous linolenic acid and palmitic acid. **E.** Synergy analysis for linolenic acid with palmitic acid for BLFX-mediated killing to ATCC 33787. Synergy is represented using a color scale or an isobologram, which compares the dose needed for 50% inhibition for synergistic agents (blue) and non-synergistic (i.e., additive) agents (red). **F.** Bliss analysis in data (E).

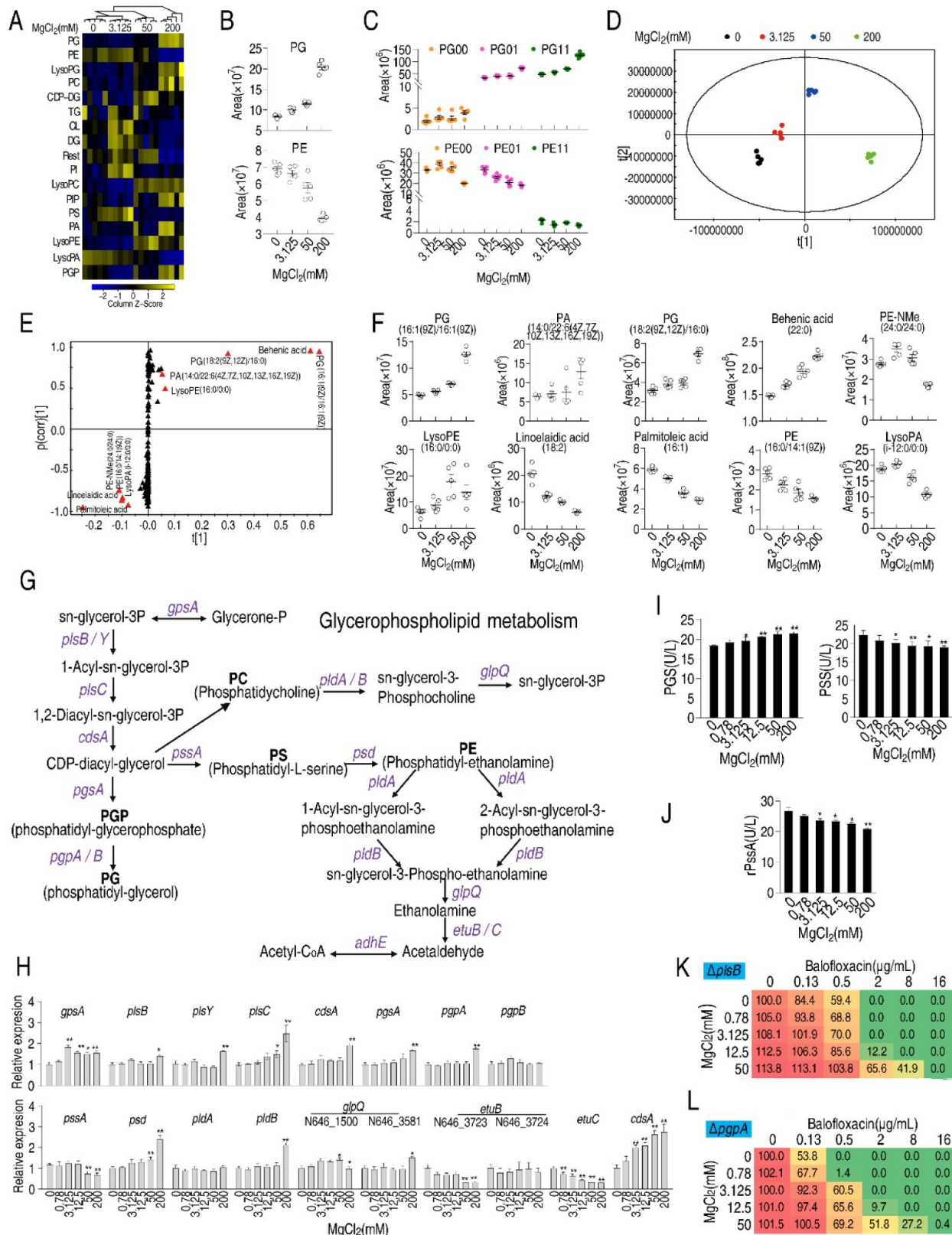


Figure 6

Effect of Mg²⁺ on phospholipid biosynthesis.

A. Heatmap showing differential abundance of lipid. **B.** Abundance of ATCC33787 phosphatidylglycerol (PG) and phosphatidylethanolamine (PE) in the indicated concentrations of MgCl₂. **C.** Scatter plots of PG and PE with different saturation in the presence of the indicated MgCl₂. **D.** PCA analysis of different concentrations of MgCl₂-induced phosphoglyceride metabolomes. Each dot represents the technical replicate of samples in the plot. **E.** S-plot generated from OPLS-DA. Predictive component *p* [1] and correlation *p* (corr) [1] differentiate 0 and 3.125 mM MgCl₂ from 50 and 200 mM MgCl₂. **F.** Scatter plot of biomarkers in data (**E**). **G.** Diagram showing phosphoglyceride metabolism. **H.** qRT-PCR for expression of genes encoding phosphoglyceride metabolism in the absence or the indicated concentrations of MgCl₂. **I.** Activity of PGS and PSS in the absence or presence of the indicated concentrations of MgCl₂. **J.** Activity of recombinant PSS in the absence or presence of the indicated concentrations of MgCl₂. **K** and **L** Synergy analysis for MgCl₂ with BLFX for Δ *plsB* (**K**) and Δ *pgpA* (**L**). Synergy is represented using a color scale or an isobologram, which compares the dose needed for 50% inhibition for synergistic agents (blue) and non-synergistic (i.e., additive) agents (red).

S-plot analysis showed that upregulated lipids include behenic acid, PG[16:1(9Z)/16:1(9Z)], PG[18:2(9Z, 12Z)/16:0], PA [14:0/22:6 (4Z, 7Z, 10Z, 13Z, 16Z, 19Z)], lysoPE(16:0/0:0) and downregulated lipids include linoelaidic acid, palmitoleic acid, PE-NMe(24:0/24:0), PE[16:0/14:1(9Z)], lysoPA(i-12:0/0:0) (**Fig 6E**). Among them, the abundance of PG[16:1(9Z)/16:1(9Z)], PG[18:2(9Z,12Z)/16:0], and behenic acid was increased, while that of linoelaidic acid, palmitoleic acid and PE[16:0/14:1(9Z)] were decreased with increasing Mg²⁺ (**Fig 6F**). Therefore, altered phospholipid abundance may play a role in the effect of Mg²⁺ on resistance to balofloxacin.

PE and PG are the two end products of phospholipid metabolism (**Fig 6G**). Expression of most genes in the phospholipid biosynthetic pathway, including *psd*, *pldB*, and *glpQ*, was increased in the presence of 200 mM MgCl₂, but not *pssA* and *etuB*, *etuC*, encoding the first enzyme and the last enzyme in the pathway, respectively. In addition, the expression of *gpsA*, encoding GpsA that transforms sn-glycerol-3P to glycerone-P, was increased in an independent of Mg²⁺ dose (**Fig 6H**). Phosphatidylglycerol phosphate synthase (PGS) encoded by *pgsA* and phosphatidylserine synthase (PSS) encoded by *pssA* are crucial enzymes for biosynthesis of PG and PE, respectively. Activities of PGS and PSS were elevated and reduced, respectively, by the increasing concentration of Mg²⁺ (**Fig 6I**). The effect of Mg²⁺ on the activity of PSS was confirmed with recombinant PSS (**Fig 6J**). The deletion of *plsB* and *pgpA*, the first and last genes for PG biosynthesis, respectively, lowered cell viability in the presence of balofloxacin (we failed to obtain Δ *psd*) (**Figs 6K-I**). These results indicate that phospholipids may play a role in the effect of Mg²⁺ on resistance to balofloxacin; more specifically, PE may inhibit and PG may promote resistance to balofloxacin.

Mg²⁺ regulates membrane polarization, permeability and fluidity to confer balofloxacin resistance

Membrane potential and membrane polarization influence membrane permeability and uptake of antibiotics (Lee *et al.*, 2019; Peng *et al.*, 2015). Thus, we speculated that exogenous Mg²⁺ reduces membrane polarization. Here, we show that 12.5-200 mM MgCl₂ promotes membrane depolarization in a dose-dependent manner (**Fig 7A**). The effect of Mg²⁺ on proton motive force (PMF) displayed MgCl₂ dose- and time- dependent manner (**Fig 7B**). Specifically, membrane depolarization decreased over time without MgCl₂. However, depolarization gradually increased for 2 h, reached a plateau for 3 - 5 h and decreased from 5 to 10 h when supplemented with 0.78 or

3.125 mM MgCl_2 . The extent of maximum depolarization and the length of time at higher level of depolarization also increased by MgCl_2 (**Figs 7C** [↗](#) and **7D** [↗](#)). These results suggest that Mg^{2+} - dependent changes in membrane depolarization may influence bacterial resistance to antibiotics.

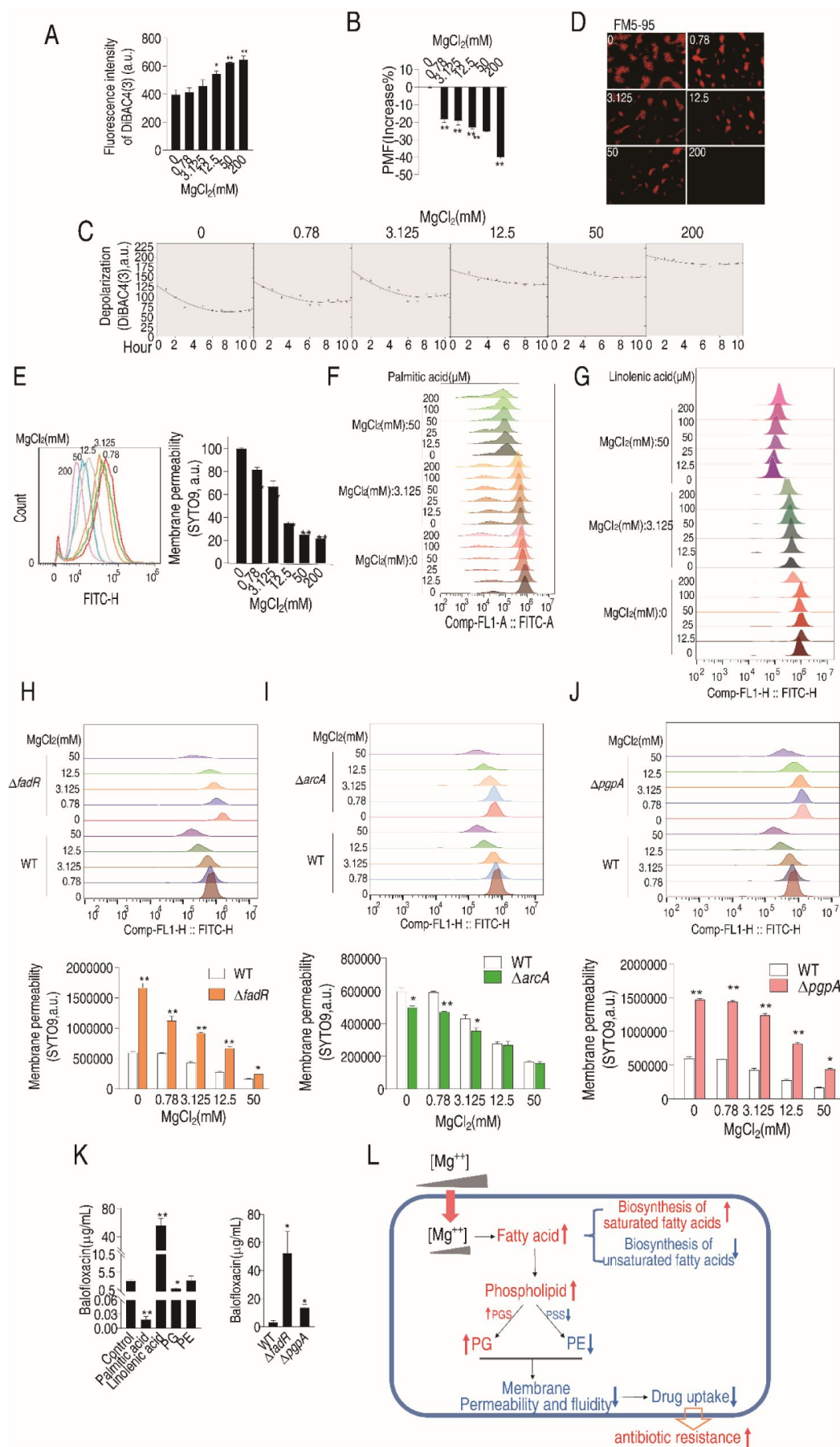


Figure 7.

Mg²⁺ regulates membrane polarization, permeability and fluidity to confer balofloxacin resistance.

A. and **B.** Depolarization (**A**) and PMF (**B**) of ATCC33787 in the absence or indicated concentrations of MgCl₂. **C** and **D.** Dynamic depolarization (**D**) Membrane fluidity of ATCC33787 in the absence or presence of the indicated concentrations of MgCl₂, detected in fluorescence microscopy. **E.** Membrane permeability of ATCC33787 in the absence or presence of the indicated concentrations of MgCl₂. **F** and **G.** Membrane permeability of ATCC33787 cultured in palmitic acid (**F**) or linolenic acid (**G**) in the indicated concentrations of MgCl₂. **H** and **I.** Membrane permeability of $\Delta fadR$ (**H**) and $\Delta arcA$ (**I**) in the absence or presence of the indicated concentrations of MgCl₂. **J.** Membrane permeability of $\Delta pgpA$ in the absence or presence of the indicated concentrations of MgCl₂. **K.** Intracellular BLFX of ATCC33787 in the presence of palmitic acid, linolenic acid, PG, and PE (left panel) or in $\Delta fadR$ and $\Delta pgpA$ mutants (right panel). **L.** Diagram for mechanisms by which Mg²⁺- mediated resistance to BLFX.

Moreover, efficacy of antibiotics is strongly influenced by bacterial membrane permeability and fluidity (Saeloh *et al.*, 2018; Zhao *et al.*, 2021). Hence, membrane permeability and fluidity were investigated with fluorescence microscopy to visualize *V. alginolyticus* cells in the presence of FM5-95 (red dye) and various concentrations of MgCl₂. Mg²⁺ decreased red staining, and no staining was observed in the presence of 200 mM Mg²⁺ (Fig 7D). Consistently, the uptake of SYTO9 dye decreased with increasing MgCl₂, indicating bacterial membrane permeability decreased in an Mg²⁺ dose-dependent manner (Fig 7E). These results indicate that exogenous Mg²⁺ decreased bacterial membrane permeability.

Furthermore, exogenous palmitic acid shifted the fluorescence signal peaks to the left in an MgCl₂- and palmitic acid concentration-dependent manner (Fig 7F). In contrast, exogenous linolenic acid shifted the peak to the right in a dose-dependent manner (Fig 7G). These results support the conclusion that exogenous palmitic acid and linolenic acid decrease or increase membrane permeability, respectively. In $\Delta fadR$ cells, membrane permeability increased and then slowly decreased with increasing Mg²⁺, remaining higher than the control (Fig 7H). However, $\Delta arcA$ cells displayed reduced membrane permeability in the presence of low but not high MgCl₂ (Fig 7I). $\Delta pgpA$ cells exhibited higher membrane permeability than control cells at all MgCl₂ concentrations, but the effect was smaller at higher MgCl₂ (Fig 7J). These data are consistent with the viability of these mutants as described above (Figs 4L and 6L).

Relative membrane permeability appeared to correlate with relative intracellular balofloxacin and antibiotic efficacy. Exogenous palmitic acid and linolenic acid reduced and promoted uptake of balofloxacin 159- and 18-fold, respectively (Fig 7K). Loss of *fadR* and *pgpA* increased intracellular balofloxacin (Fig 7K). These data suggest that bacterial membrane permeability is a critical factor in the efficacy and uptake of balofloxacin, in the presence or absence of exogenous MgCl₂.

Discussion

The present study explores the effect of Mg²⁺ on phenotypic in *V. alginolyticus*. The results suggest that exogenous Mg²⁺ promotes biosynthesis of saturated fatty acids and inhibits biosynthesis of unsaturated fatty acids, while it also upregulates the activity of PGS and the abundance of PG, and downregulates the activity of PSS and abundance of PE. As a result, membrane permeability and antibiotic uptake decrease (Fig 7L), enabling phenotypic resistance. The results are consistent with the observed impact of exogenous Mg²⁺ on bacterial survival in the presence of balofloxacin

and other antibiotics, and it is well recognized that membrane permeability critically impacts drug efficacy. Another pathway that influences drug uptake involves glutamine-inosine transport mediated by CpxA/R-OmpF (Zhao *et al.*, 2021 [DOI](#)). The results also show that exogenous fatty acids influence membrane permeability and bacterial survival in the presence of balofloxacin.

Mg²⁺ is the most abundant divalent cation in the cell (Pohland & Schneider, 2019 [DOI](#); Pontes *et al.*, 2015 [DOI](#)) and it plays many essential roles. For example, Mg²⁺ stabilizes macromolecular complexes and membranes, binds cytoplasmic nucleic acids and nucleotides, interacts with phospholipid head groups and cell surface molecules, and is an essential cofactor in many enzymatic reactions (Groisman *et al.*, 2013 [DOI](#)). A novel observation of the present study is that exogenous Mg²⁺ influences the abundance of palmitic acid and linolenic acid and that it upregulates PGS and downregulates PSS. These findings extend our understanding of the complexity of the biological functions of the divalent magnesium cation.

Recent studies suggest a link between fatty acid biosynthesis and antibiotic resistance in *E. tarda* (Su *et al.*, 2021 [DOI](#)). The present study shows that Mg²⁺ has opposite effects on the abundance of saturated and unsaturated fatty acids, stimulating biosynthesis of saturated fatty acids at moderately high levels, while inhibiting biosynthesis of unsaturated fatty acids at ≥ 3 mM Mg²⁺. Therefore, ratio of saturated and unsaturated fatty acids should be a key clue to understand antibiotic resistance.

In previous studies of gram-positive and gram-negative bacteria (Kumariya *et al.*, 2015 [DOI](#); Said *et al.*, 1987 [DOI](#)), it was shown that 10 - 20 mM Mg²⁺ disrupts *Staphylococcus aureus* membranes and kills stationary-phase *S. aureus* cells, but it does influence survival of *Escherichia coli* and *Bacillus subtilis* (Xie & Yang, 2016 [DOI](#)). Low concentrations of Mg²⁺ (≤ 10 mM) induce PmrAB-dependent modification of lipid A in wild-type *E. coli* (Herrera *et al.*, 2010 [DOI](#)). A putative zwitterionic amino-containing phospholipid increased significantly, whereas amounts of phosphatidylglycerol and cardiolipin decreased in two mutants of *Enterococcus faecium* resistant to mundticin KS (Sakayori *et al.*, 2003 [DOI](#)). However, Mg²⁺ induces antibiotic resistance through regulating the activity of PGS and PSS is not reported. More importantly, the regulation targets the two enzymes in a reverse manner to enlarge the change in the ratio between PE and PG for recognition of antibiotic resistance, which is a previously unknown mechanism mediated by Mg²⁺. Previous studies on membrane permeability also show that PhoP/Q is activated by low magnesium, and that inhibition or inactivation of PhoP/Q stimulates uptake of β -lactam antibiotics in *Stenotrophomonas maltophilia* (Huang *et al.*, 2021 [DOI](#)). The effects of the outer membrane permeabilizers, polymyxin B nonapeptide and EDTA, are completely abolished by 3 mM Mg²⁺ (Kwon & Lu, 2006 [DOI](#)). In response to Mg²⁺ limited growth, enteric Gram-negative bacteria show higher acylation of lipid A, which alters membrane permeability and reduces uptake of cationic antimicrobial peptides (Guo *et al.*, 1998 [DOI](#)). Mg²⁺ reverts the high sensitivity of rough mutants of *Salmonella typhimurium*, *S. minnesota*, and *Escherichia coli* 08 (i.e. with defects in the carbohydrate core of the lipopolysaccharide) to several antibiotics (Stan-Lotter *et al.*, 1979 [DOI](#)). Mg²⁺ (1 mM) totally inhibited aminoglycoside-mediated outer membrane permeabilization in *Pseudomonas aeruginosa* (Hancock *et al.*, 1981 [DOI](#)). However, mechanisms underlying Mg²⁺ regulation to antibiotic resistance are largely unknown. The present study reveals that Mg²⁺ plays the crucial role in the membrane permeability responsible for antibiotic resistance via modulating PE and PG in phospholipid metabolism.

In summary, the present study shows that Mg²⁺ modulates the phenotypic resistance of *V. alginolyticus* to balofloxacin and potentially other quinolones and other classes of antibiotics by a mechanism that involves altered membrane permeability and reduced antibiotic uptake. These findings inform the ongoing and future search for ways to improve the therapeutic effects of antibiotics to magnetism-induced phenotypic resistance.

Materials and Methods

Bacterial strains and culture

V. alginolyticus ATCC33787 and *V. parahaemolyticus* 01 are from our laboratory collection (Jiang *et al*, 2022 [\[4\]](#); Kou *et al*, 2022 [\[4\]](#)). They were cultured in 0.5% yeast broth (HuanKai Microbial, Guangdong, China) (pH 7.0) with 3% NaCl at 30 °C overnight. The cultures were diluted 1:50 (v/v) and grown in fresh 0.5% yeast broth supplemented with desired concentrations of MgCl₂ at 30 °C until OD₆₀₀ = 0.6. Bacteria were harvested by centrifugation at 8,000 rpm for 3 min and resuspended in corresponding concentrations of MgCl₂ to 0.6 of OD₆₀₀.

Measurement of the minimum inhibitory concentration (MIC) by microtitre- dilution-method

MIC using microtitre-dilution-method was performed as previously described (Zhang *et al*, 2019 [\[4\]](#)). In brief, the overnight bacterial cultures in 3% NaCl were diluted at 1:100 (v/v) in fresh 0.5% yeast broth supplemented with desired concentrations of MgCl₂, cultured at 30 °C and collected when the cells arrived at an OD₆₀₀ of 0.6 in medium without MgCl₂. Then, 1 × 10⁵ CFU cells were dispensed into each well of a 96-well microtiter polystyrene tray after which a series of 2-fold dilutions of antibiotic was added. Following incubation at 30 °C for 16 h, the MIC was defined as the lowest antibiotic concentration that inhibited visible growth. Three biological repeats were carried out.

Measurement of MIC by Oxford method

Determination of MIC by Oxford cup was performed as previously described (Liu *et al*, 2015 [\[4\]](#)). Bacterial cells cultured overnight in yeast medium were diluted at 1:100, shaken at 30 °C and 200 rpm until the OD₆₀₀ nm reached at 0.6. Aliquot of 100 µL cells were spread on 0.5% yeast solid medium containing 3% NaCl and 0, 0.625, 1.25, 2.5, 5, 10, 20, or 40 mM of MgCl₂. Oxford cups were placed on the solid medium, and various amounts of balofloxacin (0, 0.39, 0.78, 1.56, 3.125, 6.25, and 12.5 µg) were added. After culturing at 30 °C for 12 h, diameter of the inhibition zone was measured and the inhibition area was calculated.

Measurement of intracellular balofloxacin

Measurement of intracellular balofloxacin was performed by LC-MS analysis and plate-counting assay. For LC-MS analysis, ATCC33787 cultured in the desired concentrations of MgCl₂ were collected and adjusted to OD 0.6. Aliquot of 50 mL bacterial cells were added into a 250 mL Erlenmeyer flask and then 300 µL of 10 mg/mL balofloxacin (final concentration is 60 µg/mL). After being cultured for 6 h at 30 °C, these bacterial cells were collected and adjusted to OD 1.0 with 3% NaCl containing corresponding concentrations of MgCl₂. Aliquot of 30 mL bacteria were collected and washed three times using mobile phase (acetonitrile: double distilled water containing 0.1mol/l formic acid = 35:65) for LC-MS detection. The bacterial cells were added 1 mL mobile phase, crushed in ice water bath for 10 min (crush for 2S, pause for 3S at 35% power). Following by centrifugation, supernatants were collected, filtered, and then measured by using liquid chromatography for balofloxacin. Different concentrations of balofloxacin were used for a standard curve. For plate-counting assay, ATCC33787 cells cultured overnight were diluted at 1:100, shaken at 30 °C and 200 rpm until the OD₆₀₀ nm reached at 0.6. The cells were diluted at 1:100 and spread on LB agar to preparing ATCC33787 plates. Meanwhile, ATCC33787 cells were cultured in medium with desired MgCl₂ and at 30 °C and 200 rpm for 6 h. These cells were collected, washed and sonicated. The sonicated solution was added to Oxford cups with the ATCC33787 plates. After culturing at 30 °C for 12 h, diameter of the inhibition zone was measured. A gradient of balofloxacin was used as a standard curve for drug quantification.

Measurement of intracellular magnesium

Measurements of intracellular Mg^{2+} concentration were carried out as previously described with a modification (Yang *et al.*, 2018 [↗](#)). In brief, *V. alginolyticus* cultured in 0%, 0.78%, 3.125%, 12.5%, 50%, and 200% $MgCl_2$ were collected, washed, and resuspended in the same concentrations of $MgCl_2$ to an OD600 of 1.0. Aliquots of 20 mL of bacterial suspensions were centrifuged and washed by ddH₂O once. The resulting cells were weighted, which was designated wet weight. These samples were freeze-dried overnight, which was designated dry weight. Intracellular water volume was calculated using the following formula: $W \times (1 - W/D - 0.23)$ as described previously (Unemoto *et al.*, 1973 [↗](#)). Two hundred microliters of concentrated nitric acid was added to the dried cells and then heated in 75 °C for 20 min. After diluting the samples 20- fold, they were analyzed for Mg^{2+} concentration by inductively coupled plasma mass spectrometry (ICP-MS) (iCAP 6500, Thermo Fisher). The intracellular Mg^{2+} concentration was calculated according to the volume of intracellular water.

Western blotting

Western blotting was carried out as described previously (Yao *et al.*, 2019 [↗](#)). Bacterial protein samples were prepared by ultrasound treatment, resolved on a 12% SDS-PAGE and transferred to nitrocellulose membranes (GE Healthcare Life Sciences). The membranes were incubated with 1:100 of the primary mouse antibodies, followed by goat anti-mouse secondary antibodies conjugated with horseradish peroxidase. Band intensities were detected by using a chemiluminescence imaging analysis system, Tanon-5200.

Metabolomics analysis

Metabolomics analysis was performed by GC-MS as described previously (Yang *et al.*, 2018 [↗](#)). Briefly, ATCC33787 were cultured in 0.5% yeast broth with desired $MgCl_2$. Equivalent numbers of cells were quenched with 60% (v/v) cold methanol (Sigma) and then centrifuged at 8,000 rpm at 4 °C for 5 min. One milliliter of cold methanol was used to extract metabolites. To do this, the samples were sonicated for 5 min at a 10- Wpower setting using the Ultrasonic Processor (JY92-IIDN, Scientz, China), followed by centrifugation at 12,000 rpm in 4 °C for 10 min. Supernatants were collected and 10 μ L ribitol (0.1 mg per mL, Sigma-Aldrich, USA) was added into each sample as an internal quantitative standard. The supernatants were concentrated for metabolite derivatization and then used for GC-MS analysis. Every experiment was repeated by five biological replicates. GC-MS detection and spectral processing for GC-MS were carried out using the Agilent 7890A GC equipped with an Agilent 5975C VL MSD detector (Agilent Technologies, USA) as described previously (Unemoto *et al.*, 1973 [↗](#); Yang *et al.*, 2018 [↗](#)). Statistical difference was obtained by Kruskal Wallis test and Manne Whitney test using SPSS 13.0 and a *p* value < 0.01 was considered significant.

Hierarchical clustering was completed in the R platform (<https://cran.r-project.org/> [↗](#)) with the function “heatmap. 2” of “gplots library”. Z score analysis was used to scale each metabolite. Multivariate statistical analysis included principal component analysis (PCA) and orthogonal partial least square discriminant analysis (OPLS-DA) implemented with SIMCA 12.0 (Umetrics, Umeå, Sweden). Control scaling was selected prior to fitting. All variables were mean centered and scaled to pareto variance of each variable. PCA was used to reduce the high dimension of the data set. Differential metabolites to their respective biochemical pathways were outlined in the MetaboAnalyst 3.0 (<http://www.metaboanalyst.ca/> [↗](#)). Pathways were enriched by raw *p* value < 0.05.

qRT-PCR

Quantitative real time polymerase chain reaction (qRT-PCR) was carried out as described previously (Yang *et al.*, 2020 [\[1\]](#)). Total RNA was extracted from *V. alginolyticus* using TRIZOL reagent (Invitrogen Life Technologies) according to the protocol. RNA Electrophoresis was carried out in 1% (w/v) agarose gels to identify quality of the extracted RNA. By using a PrimeScript™ RT reagent Kit with gDNA eraser (Takara, Japan), reverse transcription-PCR was carried out on 1 µg of total RNA and primers are listed in **Table EV2**. qRT-PCR was performed in 384-well plates with a total volume of 10 µL and the reaction mixtures were run on a LightCycler 480 system (Roche, Germany). Data are shown as the relative mRNA expression compared to 0% MgCl₂ test with the endogenous reference 16S rRNA gene.

Antibiotic bactericidal assay

Antibiotic bactericidal assay was performed as described previously with a modification (Li *et al.*, 2016 [\[2\]](#); Zhao *et al.*, 2021 [\[3\]](#)). The cultured bacteria of ATCC33787 and its mutant strains were transferred to fresh yeast medium at a dilution of 1:1000 and dispensed into test tubes and then the indicated concentrations of MgCl₂ and balofloxacin were added. If desired, 2 mM 2-aminooxazole and 1 µg/mL triclosan were or a metabolite complemented. These mixtures were incubated at 30 °C and 200 rpm for 6 h. Cells were collected. To determine CFU per mL, 100-µL samples were 10-fold serially diluted with 900 µL M9 buffer and an aliquot of 5 µL of each dilution was spotted onto the LB agar plates and cultured at 30 °C for 8 h.

Construction of gene-deleted mutants

Construction of gene-deleted mutants was carried out as described previously (Kuang *et al.*, 2021 [\[4\]](#)). Primers were designed as shown in **Table EV3** using CE Design V1.03 software. To construct gene-deleted mutants, upstream and downstream 500-bp fragments were amplified from the genome using two pairs of primers (primers P1 and P2, primers P3 and P4), and then merged into a 1,000-bp fragment by overlap PCR using a pair of primers (primers P1 and P4). After the fragments were digested by *Xba*I, they were ligated into the pDS132 vector digested by the same enzymes, transformed into MC1061 competent cells. Conversion products were coated on LB plate containing 25 µg/mL chloramphenicol and cultured at 37 °C overnight. The plasmids from colony growing on the plate were identified by PCR using a pair of primers (primers P1 and P4) and sequenced. The sequenced plasmids were transformed into MFD λ pir competent cells as donor. MFD λ pir and recipient bacterium ATCC 33787 were cultured to an optical density (OD) of 1.0 and then mixed at a ratio of 4:1. After centrifugation, the pellets were resuspended with 50 µL LB medium including DAP (100 µg/mL), dropped onto sterilized filter paper on LB medium including DAP (100 µg/mL), and cultured for 16-18 h at 37 °C.

All bacteria rinsed from the filter paper with LB medium were smeared onto the LB plate with chloramphenicol (25 µg/mL) and Ampicillin (100 µg/mL). After cultured for 16-18 h at 37 °C, bacteria were screened by LB plate with above two antibiotics. The bacteria were identified by plasmid PCR using a pair of primers P1 and P4 and sequencing and then were cultured and smeared onto the LB plates with 20% sucrose. The clones were cultured and smeared onto the LB plates with 20% sucrose again. The clones, which did not grow on the LB plates with chloramphenicol but grew on the LB plates with 20% sucrose, were identified by PCR using primer P7P8 (primer P7 is set at about 250 bp upstream of P1, and primer P8 is set at about 250 bp downstream of P4), P4P7, and P5P6 (for amplification of the target gene).

LC-MS analysis for lipidomics

ATCC33787 were cultured in medium with desired concentrations of MgCl₂, washed by 3% NaCl with the desired concentrations of MgCl₂, and adjusted to OD 1.0. Aliquot of 30 mL the cultures was centrifuged each sample and bacterial cells were collected. Aliquot of 800 µL distilled water

was added and treated for 5 min at 100 °C to inactivate phospholipase C. Protein concentration was determined with BCA kit. Sample solution with 3 mg protein was transferred into a 10 mL centrifuge tube, and supplemented with distilled water to 800 µL. Then all samples were processed as follows: 1 mL chloroform and 2 mL methanol (to make the volume ratio of chloroform: methanol: water = 1:2 : 0.8) were added and cholic acid was as an internal standard; These mixtures were vortexed for 2 min and then 1 mL chloroform was add and vortex for 30s; 1 mL 10% NaCl solution was add, vortexed for 30s, and placed at room temperature overnight. After the solution was layered, a 2.5 mL syringe was used to suck the lowest layer (chloroform layer) to a new EP tube. Solution of the chloroform layer was evaporated in the rotary evaporator for 2 h and then 1 mL mobile phase (50% and 50%B. A, 25 mM ammonium acetate / methanol (30 : 70); B, methanol) was added for analysis of lipids by LC-MS. Database (<https://hmdb.ca>) was used for lipid identification.

Measurement of enzyme activity

Enzyme activity was carried out as previously described (Jiang *et al*, 2020). Cells cultured in medium were collected and washed three times with saline. The bacterial cells were suspended in Tris-HCl (pH 7.4) and disrupted by sonic oscillation for 6 min (200 W total power with 35% output, 2 s pulse, 3 s pause over ice). After centrifugation, supernatants were collected. The protein concentration of the supernatant was determined using Bradford assay (Beyotime, P0009). Then, 200 µg proteins were used for determination of pyruvate dehydrogenase (PDH), α-ketoglutarate dehydrogenase (KGDH) and succinate dehydrogenase (SDH) activity.

Measurement of proton motive force

Measurement of the transmembrane voltage delta PSI (δPSI), which is the electrical component of the PMF, was performed as previously described (Cheng *et al*, 2018). Bacteria cultured in medium with desired concentrations of MgCl₂ were collected at centrifugation and labeled by DiO2(3). Approximately 1×10^7 CFU were added to a flow cytometry tube containing 1 mL buffer with 10 µM DiOC2(3) (Sigma) and incubated in the dark for 30 min at 30 °C. Samples were assayed with BD FACSCalibur flow cytometer with a 488 nm excitation wavelength. Gates for bacterial populations were based on the control population by using forward versus side scatter and red versus green emission. Size and membrane potential determined the intensity of Red (488 nm excitation, 610 nm emission) fluorescence. The diverse ratios of red and green indicated fluorescence intensity values of the gated populations. Computational formula of membrane potential:

$$\text{Log}(10^{3/2} \times (\frac{\text{red fluorescence}}{\text{green fluorescence}}))$$

Measurement of depolarization

The membrane potential was estimated by measuring the fluorescence of the potential- dependent probe DiBAC4(3) (Saint-Ruf *et al*, 2016). ATCC33787 were cultured in medium with desired concentrations of MgCl₂, collected and then adjusted to OD 0.6. Aliquot of 100 µL bacterial cells were diluted to 1 mL, and 2 µL of 5 mM DiBAC4(3) was added. After incubated for 15 min at 30 °C without light and vibration, these samples were filtered and detected by BD FACSCalibur.

Measurement of fluidity by fluorescence microscopy

ATCC33787 were cultured in medium with desired concentrations of MgCl₂, collected and then adjusted to OD 0.6. Aliquot of 100 µL bacteria cells of each sample were diluted to 1 mL and 10 µL (10 mg/mL) FM5-95 was added. After incubated for 20 min at 30 °C at vibration without light, the

sample was centrifuged for 10 min at 12,000 rpm. The pellets were resuspended with 20 μ L of 3% NaCl. Aliquot of 2 μ L sample was dropped on the agarose slide, and take photos under the inverted fluorescence microscope.

Measurement of membrane permeability

Measurement of membrane permeability was carried out as described previously (Su *et al.*, 2021 [DOI](#)). ATCC33787 were cultured in medium with desired concentrations of MgCl_2 , collected and then adjusted to OD 0.6. Aliquot of 100 μ L bacteria cells of each sample were diluted to 1 mL and 2 μ L 10 mg/mL SYTO9 was added. After incubated for 15 min at 30 °C at vibration without light, the mixtures were filtered and measured by BD FACSCalibur.

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Author Contributions

- Peng conceived the idea, designed the experiments and supervise the projects. H. Li and J. Yang, S.F. Kuang conducted the experiments, and interpreted the results.

Declaration of interest statement

Authors declare that they have no competing interests.

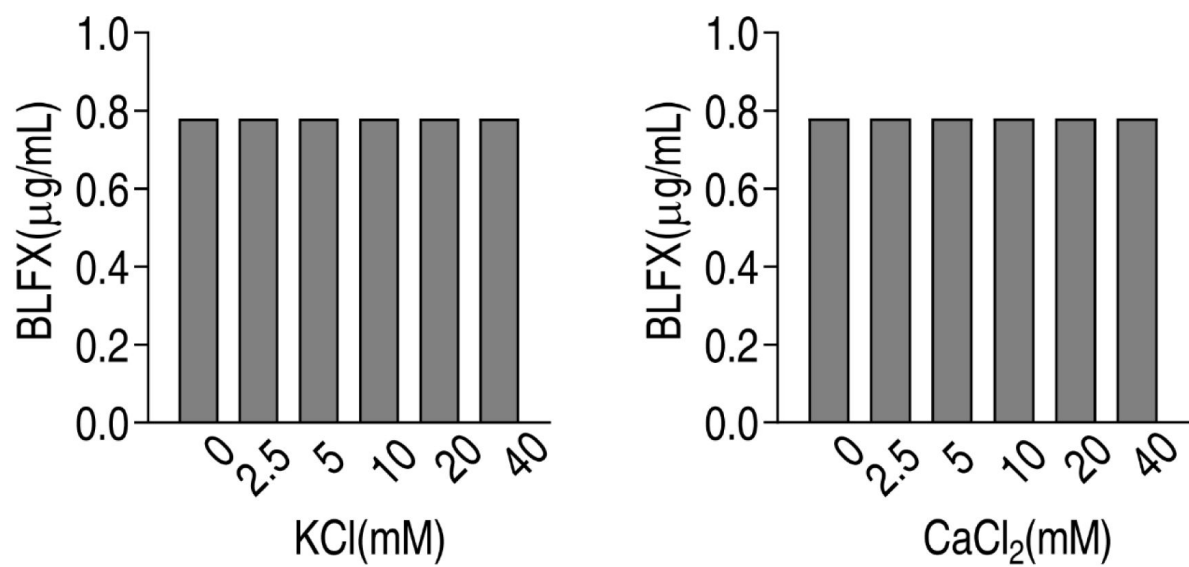


Figure EV1

MIC of ATCC33787 to BLFX in the absence or presence of the indicated concentrations of KCl or CaCl₂.

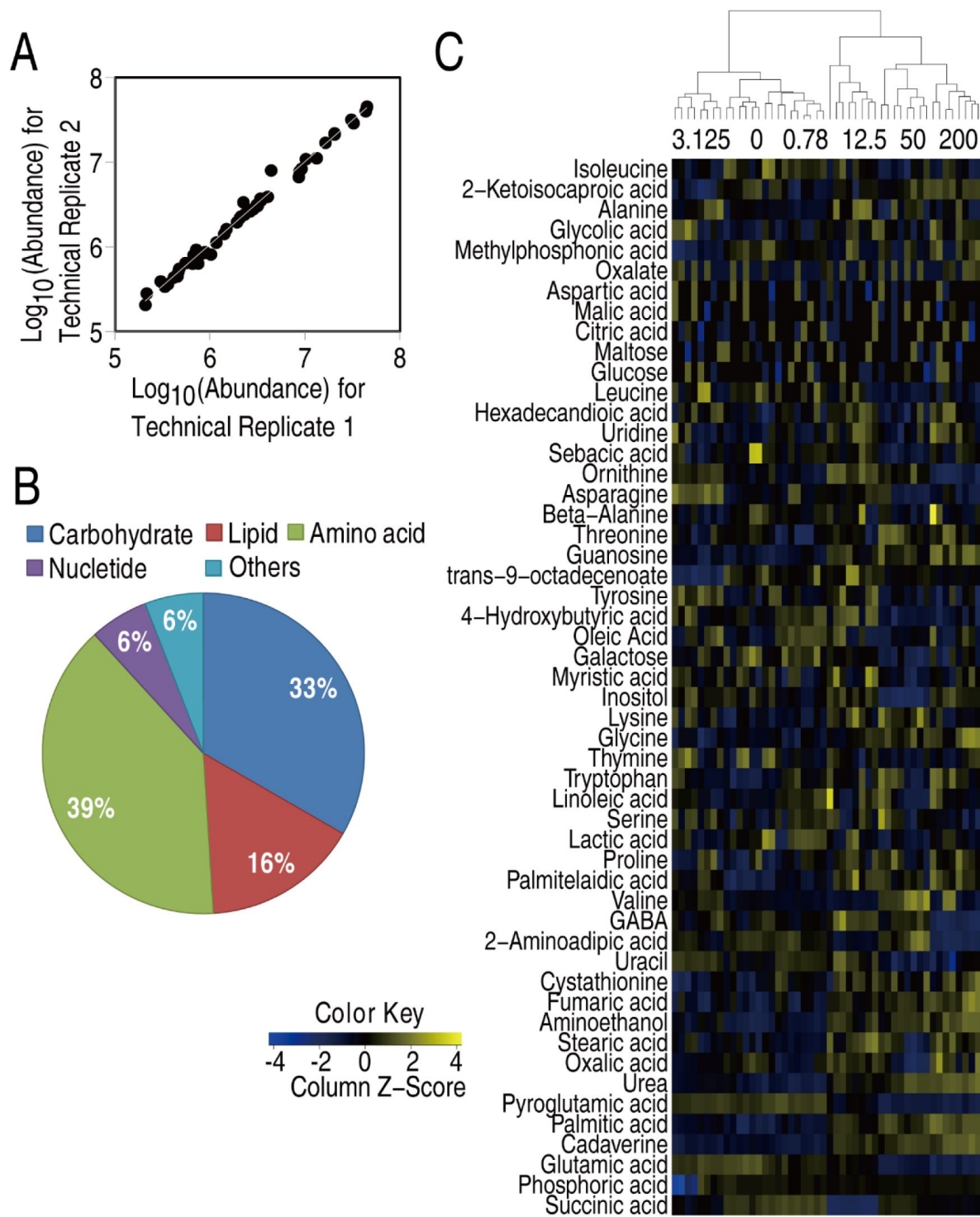


Fig EV2

Metabolic profiles of *V. alginolyticus* in different concentrations of MgCl_2 .

- A.** Reproducibility
B. Percentage of metabolites in every category
C. Heatmap of metabolites

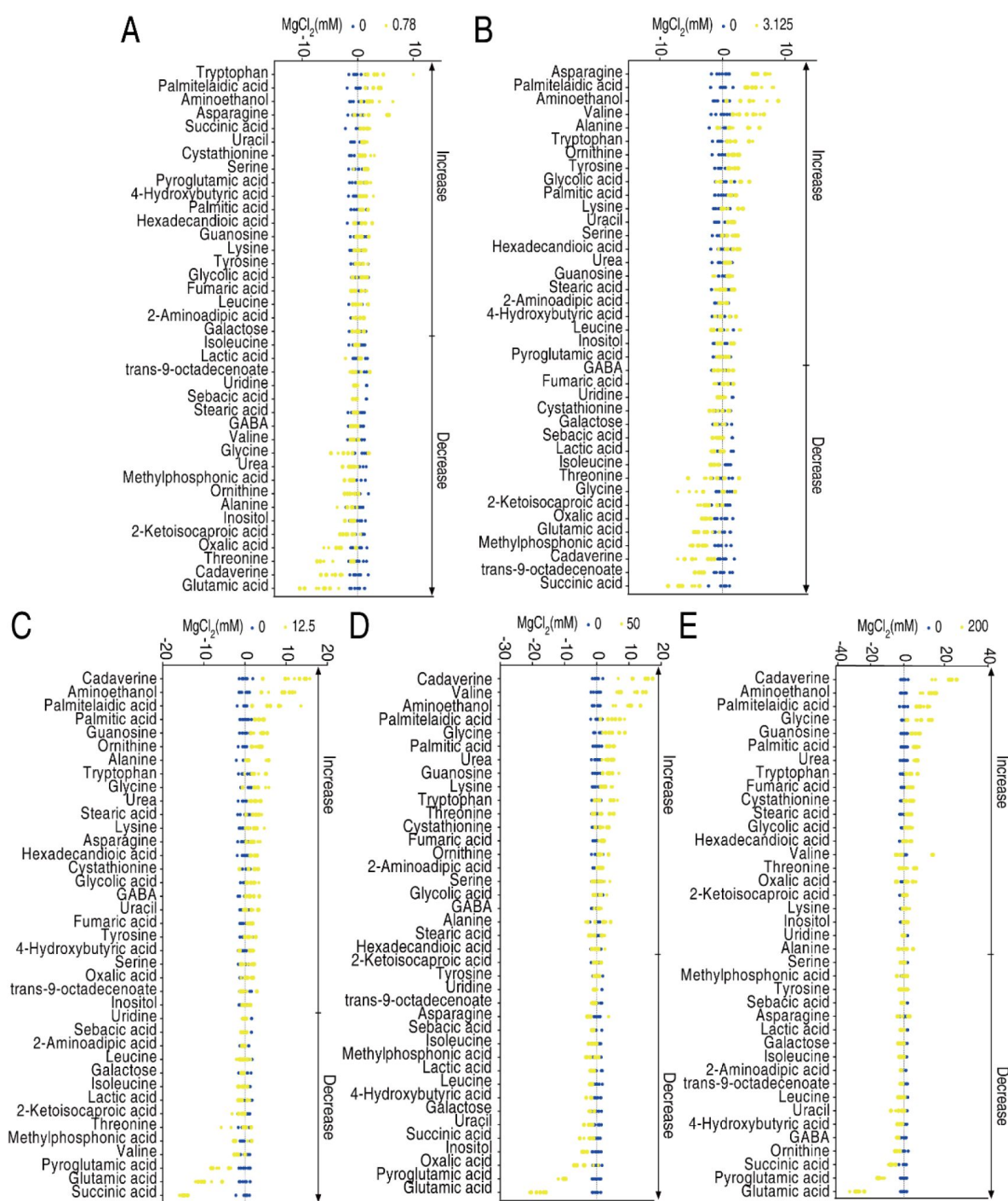


Fig EV3

Heatmap and Z score plots of differential metabolites.

- A.** Heatmap of differential metabolites
B-F. Z score plots of differential metabolites.

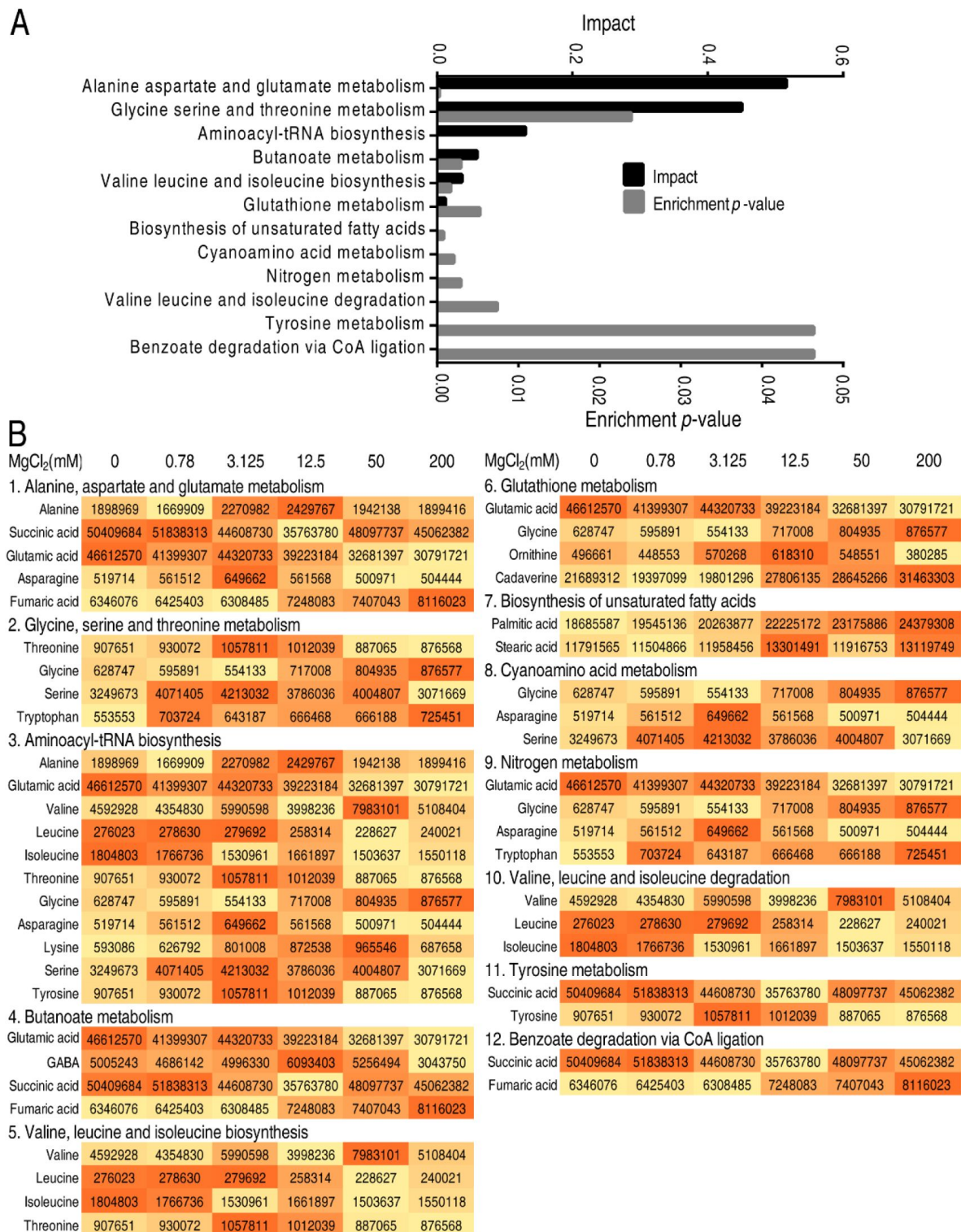


Fig EV4

Pathway enrichment of differential metabolites.

- A.** Pathway enrichment of differential metabolites
B. Differential metabolites in enriched pathways

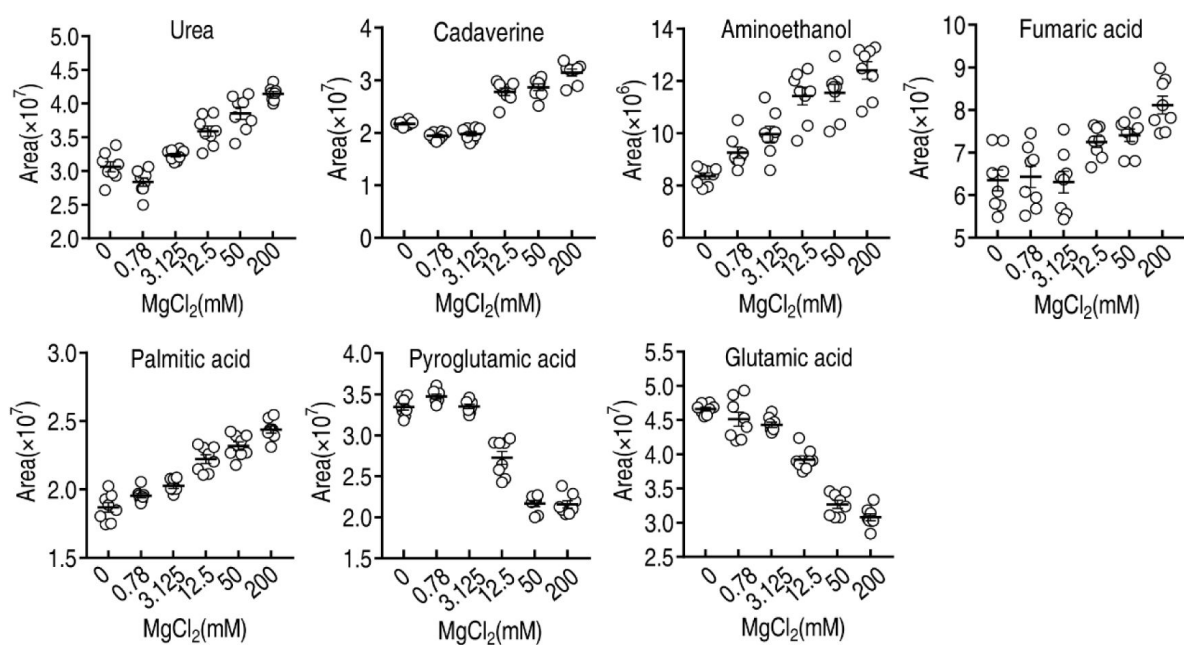


Fig EV5

Scatter plots of differential metabolites identified by S-plot.

Fig EV6

Lipidomes in the different concentrations of MgCl_2 .

- A. Area of fatty acids in the presence of indicated concentrations of MgCl_2 .
B. Percentage of lipids, saturated fatty acid and unsaturated fatty acid in the presence of indicated concentration of MgCl_2 .
C. Volcano plots of lipidomics of indicated concentration of MgCl_2 as compared to non-treated control.
D. Relative abundance of saturated fatty acids, unsaturated fatty acids and lipids in the presence of indicated concentrations of MgCl_2 .
E. Relative percentage of indicated lipids.

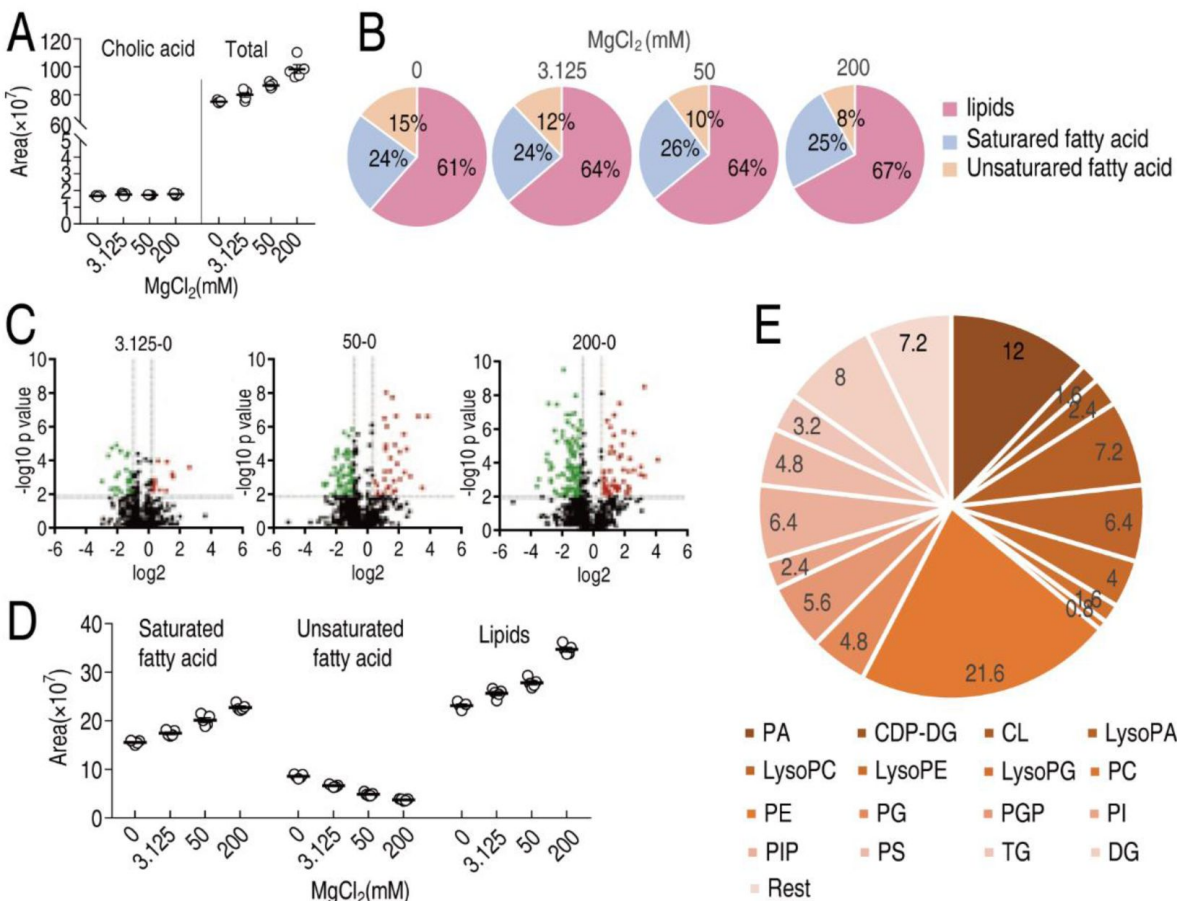


Table EV1

Comparison in components in LBS and SWT

	LBS	SWT
Tryptone(%)	1	1
Yeast extract(%)	0.5	0.5
NaCl(mM)	171	210
MgSO ₄ (mM)	0	35
KCl(mM)	0	7
CaCl ₂ (mM)	0	7

Table EV2

Primes used in the present study

gene		KEGG entry	Sequences (5'-3')	gene		KEGG entry	Sequences (5'-3')
<i>accA</i>	Forward	N646_1411	AAAGGTCGTGAAACCAAAGA	<i>fadD</i>	Forward	N646_4025	CATTATTACTGCGATCCTCG
	Reverse		AGCCGTGTCGATAAATGTG		Reverse		CCATATTGAGATATTTGTCTGGG
<i>accB</i>	Forward	N646_1967	TCAAGCACCAATTCACACG	<i>fadE</i>	Forward	N646_1395	TATCCCAACTGATGTCAAAGG
	Reverse		ACTTCTGACCCACTTCTACG		Reverse		CAACCGACAGACACTCAACC
<i>accC</i>	Forward	N646_1968	AGAAGTGACTGGTGTGTGG	<i>fadB</i>	Forward	N646_2210	ATCACTAAAACGCCAGAAAA
	Reverse		GGATAGTGTCTGCCTTTGGAC		Reverse		GCTGAAGCCGCCAAAGTA
<i>accD</i>	Forward	N646_1292	CGAACTTGAGCCACAGGATA	<i>fadA</i>	Forward	N646_1311	CAGATTAGCGGCCAGTTTCT
	Reverse		ACGACTGAACCCATTGAACC		Reverse		TCAAGCGGCGGTTAATGT
<i>accD</i>	Forward	N646_4704	AAGTTTGTCCGAGCAGTTGA	<i>fadA</i>	Forward	N646_2209	ATTGCGGGCTGCGATC
	Reverse		GAGACGCCACCATAAGTTTG		Reverse		GCACACGGCAGAGACTGC
<i>fabD</i>	Forward	N646_1131	CTCGCTGGTCAAGTCGT	<i>fadH</i>	Forward	N646_1258	AAGTGCTGCCCTGCTCG
	Reverse		TGCTGGCTTCATAAGTGCA		Reverse		ATGATGGCGTCGAGTTAATG
<i>fabH</i>	Forward	N646_1132	GACCGTTCAACCATCATCCT	<i>fabR</i>	Forward	N646_2038	TTTGCTACTACGCCAACTGATGCGT
	Reverse		GCCATTTATCCGCATCTTG		Reverse		CGCTCACGAACAACAACAGAA
<i>fabH</i>	Forward	N646_1439	GACGCTCCAACATGGCTAG	<i>fabR</i>	Forward	N646_1147	AATGCTAGGCGCTGGAA
	Reverse		AGTTCGCTGGTGACAATAAGA		Reverse		TGAATGCGGCTTCCTG
<i>fabH</i>	Forward	N646_4350	GGAAGACAACGGTTACTGGG	<i>arcA</i>	Forward	N646_1744	TGGGTTGAATGGTTAGTGATG
	Reverse		TGCGGAATCACAACATCG		Reverse		TGAAGCCAGTGACGGTGAA
<i>fabB</i>	Forward	N646_1292	TTGACAGTGCGCCCATC	<i>crp</i>	Forward	N646_1004	AAGGTTGCTGCTACTTGCTGA
	Reverse		TGTCTTGCTACGCCATTTAAGA		Reverse		GCTTCGGATTGGTAATGGTT
<i>fabB</i>	Forward	N646_3036	CAACATTAATGCTGCGGCT	<i>crp</i>	Forward	N646_1885	CAAAACAACCAGACGCAATG
	Reverse		TGAGCGGCCACATTAC		Reverse		CTCACGAGAACAGCCAACAAT
<i>fabF</i>	Forward	N646_1128	GGTGTGCTATCGGCTCA	<i>gpsA</i>	Forward	N646_1922	GTGCGGGTATGTCTGATGG
	Reverse		ATGTTTACAATGGTCGAAGGA		Reverse		ATAAAGGTTTCTGGCTGTGC
<i>fabF</i>	Forward	N646_3033	AACTGCCAAAGCACTACAAGC	<i>plsB</i>	Forward	N646_2043	GCTTAGTTTCGCTTTAGTCCC
	Reverse		TGATGAGCCGTATGCGATAC		Reverse		CACCTGGCGTTGTGGC
<i>fabG</i>	Forward	N646_1130	CAACCTAACGCCAATCTACC	<i>plsY</i>	Forward	N646_2566	CGCCAAGAATGATAGGGTC
	Reverse		CCACGAGACGCAACTTCA		Reverse		TTATTAGGTTTCGGTTTCCAGT
<i>fabG</i>	Forward	N646_3034	GACCGAAGAAGAATGGGATG	<i>plsC</i>	Forward	N646_1749	CGATTTCGCTCGGTCAA
	Reverse		TGCGCTGTAGTTAGTTTGACC		Reverse		TACTGTTTCTGCGGCTGT
<i>fabG</i>	Forward	N646_3826	AGTAAAGGCGGCGTCATC	<i>cdsA</i>	Forward	N646_1423	TTGCTGCCCTTGTGCC
	Reverse		TCTGGCAGGTCTTTGGTCA		Reverse		GCCAAGATTTCGTCTGTTTT
<i>fabG</i>	Forward	N646_4505	CCCAAGGCGTTATCATCA	<i>pgsA</i>	Forward	N646_1026	GTTTTGGCGCGTTTATT
	Reverse		TTCGGGTTTCATTTGCTCT		Reverse		TTTCTCGCGCAATCATC
<i>fabZ</i>	Forward	N646_1416	GAGAAGTACCTTCACGCGAT	<i>pgpA</i>	Forward	N646_2838	ATGAGTTTGCGGGCTTT
	Reverse		CTGTTGCCTGAGCCATTG		Reverse		TCCCGCCACGATATCAT
<i>fabV</i>	Forward	N646_4267	AGTAATGGGCGGCGAAG	<i>pgpB</i>	Forward	N646_3562	TTTCTTTGGCGGGTTGT
	Reverse		TGGGTGCGTAATGTCAGG		Reverse		CTTCAGGGCGGTGCAT
<i>fabV</i>	Forward	N646_0267	GGTGGTTTACTCACTGGCATC	<i>pssA</i>	Forward	N646_2008	GATAAAACCGCCAACGA
	Reverse		TCTTCACCGCCATTACC		Reverse		TTACGTGCGGCGATATT
<i>tesA</i>	Forward	N646_4268	AGCATTTCTGGCGATACAAC	<i>psd</i>	Forward	N646_1916	CACTCTGCGCCAAATG
	Reverse		GGAGGGAAACCTCTTAGGC		Reverse		ACGAGGCGGAGTGATC
<i>tesB</i>	Forward	N646_0021	CTTTGGATTCTGTCTAATGG	<i>pldA</i>	Forward	N646_4183	CTTTTTAGGTCGCCAGAAC
	Reverse		ATAGAGTGGTCAATGGTGGC		Reverse		CATAACCCGCTGAGTATTGT
<i>yciA</i>	Forward	N646_1052	TGTCTCAGCTCGACTTAGCG	<i>pldB</i>	Forward	N646_2069	GATCTCTACCGCCAAGGT
	Reverse		GGGTTTGACCCATACTTCCA		Reverse		AGTGCGCGATGATGAAG
<i>fadL</i>	Forward	N646_1315	CTAAGTATTGGCGCAACGTACG	<i>glpQ</i>	Forward	N646_1500	CTATCCAACCCGATTCCC
	Reverse		GCCGCCGACTTGTTCTG		Reverse		ACCAAGGCGCTTTAATTT

<i>fadL</i>	Forward		CTGCCTTCGCCGATGAT	<i>glpQ</i>	Forward		AAAAGTCCAATGCCGCA
	Reverse	N646_1316	GAAATTCGCGATTGGTTTG		Reverse	N646_3581	GTAACCGGCTTGTGGAG
<i>fadL</i>	Forward		TTATCTGCGGGTGCCTCT	<i>glpQ</i>	Forward	N646_3723	CACAAATGGCGCTATCTG
	Reverse	N646_3657	GTCACGCCCGTCACCAT		Reverse		CCATCGGCTTCTTCACTC
<i>fadD</i>	Forward		TGTTCTTGAGCTGGTGGAAA	<i>glpQ</i>	Forward	N646_3724	GACTTGTGAAGCCCGC
	Reverse	N646_1045	AACACGCAATGACGAGGTAT		Reverse		AGATACTCAGGGCCAATAAAC
<i>fadD</i>	Forward		GCATCGGTCGATTCTTCC	<i>etuC</i>	Forward	N646_3725	CTTTTAAGCGCTGATGATGA
	Reverse	N646_2540	CGCCAATCTTAACCTGTGC		Reverse		GTTAGACCGGGTCTCTCTC
<i>fadD</i>	Forward		CTCTGATGATGCCAAACCTAC	<i>adhE</i>	Forward	N646_1240	AGCGATGCTGCACGAA
	Reverse	N646_3023	ACGAGGAGTGTAAGTGGGTT		Reverse		CCTGTAACTAATATGGCTGAAC

Table EV2 (continued)

<i>gene</i>		primer (5'-3')
<i>fadR</i>	P1	aaaaaggatcgatcct TACTTCCCCTGAACTCAAGCAC
	P2	TGTCAGCGCTTATCTACTTTTTATCCAATATTGATTTCTG
	P3	ATAAAAAGTAGATAAGCGCTGACAAGTTGGATAAAC
	P4	atcgcatcggtacc GCAAAGAGATGACTTGCATAATGAGTA
	P5	ATGGTCATAAAGGCGAAAAGCC
	P6	TTAGCTGTCGTCTTCCGTGAAGTT
	P7	AATAGTGCTTAAGTATCTGTTTGT
	P8	ATGACCATAGTGTTGCAAGA
<i>arcA</i>	P1	aaaaaggatcgatcct GACCTCAAGTTACTGTATGGGTTC
	P2	TAGCGGTTAAATTGGCGGTACCTAAATTTGTGACA
	P3	TAGGTACCGCCATAGATTTAACCGCTAATATGCAAAGGG
	P4	atcgcatcggtacc ATTGATGGCAGGCTGTAGTTCC
	P5	ATGCAAACCCCGCAGATTCT
	P6	TGGCGATGATTTCTGGCGT
	P7	CGACGGCAAAGCTTTCCTC
	P8	AAAAGTTGGTTGGAACAAAGGAA
<i>plsB</i>	P1	aaaaaggatcgatcct TTAGGTGTTCTCTGCGGCAT
	P2	AAGGTAAAACGGAAAGTTCACAGCCTCAATAGGTATTC
	P3	GCTGTGAACCTTTAACC GTTTTACCTTCTTCCAATAAGA
	P4	atcgcatcggtacc AGCTATTTT TAGGCTTGGCTTTTA
	P5	ACAACGATAATGATGTCCCAACC
	P6	GAACGACCACCCTCACTGAAGTA
	P7	GCTCATCATTGGCTGCGTCT
	P8	TGCTGACCATGAACCCGC
<i>waaf</i>	P1	aaaaaggatcgatcct AGTTGTGGGTTGCTGTGTTGA
	P2	GAAC TAACATGAGCTTTGTTTTTCCAGTATAGGAATT
	P3	GAAAAACAAAGCTCATGT TAGTTCGTATTGTTTATACCTT
	P4	atcgcatcggtacc TTGGCGTAGTTCTGACACGATT
	P5	AACGCCATCCAGATAGCCAAAT
	P6	AGGTTTCAAGCCCGCCAAGA
	P7	ATATGTGGGTACTACAGTTGCTGCA
	P8	CATCTAATACCTGTTCTGTCCTC
<i>lpxA</i>	P1	aaaaaggatcgatcct TACGACCTGTTGCCACGTTTG
	P2	ATTCGCCAATTAATTA AAACTCTCGACGAGCACAC
	P3	GAGAGTTTTAATTAATTGGCGAATGAACCAATAAATAA
	P4	atcgcatcggtacc ATTTTAAAGATGCGCTTCTGACG
	P5	GGCAGCGGTGGTAGAAGAGG
	P6	AACGGGTGAATCGCAGACAG
	P7	ATTGATGAGATGAATAAGCGCCT
	P8	TAACTCGCTACCACGACTGCCT
<i>lpxC</i>	P1	aaaaaggatcgatcct AGACGGTTGGTAATACAGTGAAG

Table EV3

Primers used in the present study for construction of gene-deleted mutants

<i>pgpA</i>	P2	GCCAAGTATTTACTGCTTGCCTCTCAAAAATTTGC
	P3	AGAGGCAAGCAGTAAATACTTGGCATTAGAATTAATAA
	P4	atcgcgatcggtacc CTGGAGAAATCTTACAGCTTAAC
	P5	ATTGAATCAGACGAGCAACG
	P6	CGAATGTAGTCCACTCCCAC
	P7	AGAATGCCGTTCAAGGTATTGC
	P8	TTTACTTAAAGTGGTGAAGTATT
	P1	aaaaaggatcgatcct AAGCACTAGAGCGAGCTCATTAC
	P2	GTTTCATGGTTTAATCCTTAAACCTATACTGCTAGTTT
	P3	AGGTTTAAGGATTAAACCATGAACTAAACACTGGACGT
	P4	atcgcgatcggtacc ATACCCTCGTGGTAGCGGCAT
	P5	GGCACAATTACCATTACGG
	P6	TAGAAACAGGGCAACACCAG
	P7	CATTTTTCGAACTGGCTGATT
	P8	GCAACGTGGTTATGACCAATT

Small letters are homologous recombination sequences with pDS132 plasmid.

Table EV3 (continued)

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Editors

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Reviewer #1 (Public Review):

Summary:

In the manuscript entitled "Magnesium modulates phospholipid metabolism to promote bacterial phenotypic resistance to antibiotics", Li et al demonstrated the role of magnesium in promoting phenotypic resistance in *V. alginolyticus*. Using standard microbiological and metabolomic techniques, the authors have shown the significance of fatty acid biosynthesis pathway behind the resistance mechanism. This study is significant as it sheds light on the

role of an exogenous factor in altering membrane composition, polarization, and fluidity which ultimately leads to antimicrobial resistance.

Strengths:

- (1) The experiments were carried out methodically and logically.
- (2) An adequate number of replicates were used for the experiments.

Weaknesses:

- (1) The introduction section needs to be more informative and to the point.
- (2) The weakest point of this paper is in the logistics through the results section. The way authors represented the figures and interpreted them in the results section (or the figure legends) does not match. The figures are difficult to interpret and are not at all self-explanatory.
- (3) There are too many mislabeling of the figure panels in the main text which makes it difficult to find out which figures the authors are explaining. There should be more explanation on why and how they did the experiments and how the results were interpreted.

<https://doi.org/10.7554/eLife.100427.1.sa2>

Reviewer #2 (Public Review):

Summary:

In this study, the authors aimed to identify if and how magnesium affects the ability of two particular bacteria species to resist the action of antibiotics. In my view, the authors succeeded in their goals and presented a compelling study that will have important implications for the antibiotic resistance research community. Since metals like magnesium are present in all lab media compositions and are present in the host, the data presented in this study certainly will inspire additional research by the community. These could include research into whether other types of metals also induce multi-drug resistance, whether this phenomenon can be observed in other bacterial species, especially pathogenic species that cause clinical disease, and whether the underlying molecular determinants (i.e. enzymes) of metal-induced phenotypic resistance could be new antimicrobial drug targets themselves.

Strengths:

This study's strengths include that the authors used a variety of methodologies, all of which point to a clear effect of exogenous Mg^{2+} on drug resistance in the targeted species. I also commend the authors for carrying out a comprehensive study, spanning evaluation of whole cell phenotypes, metabolic pathways, genetic manipulation, to enzyme activity level evaluation. The fact that the authors uncovered a molecular mechanism underlying Mg^{2+} -induced phenotypic resistance is particularly important as the key proteins should be studied further.

Weaknesses:

I believe there are weaknesses in the manuscript, however. The authors take for granted that the reader is familiar with all the assays utilized, and do not properly explain some experiments, and thus I highly suggest that the authors add a brief statement in each situation describing the rationale for each selected methodology (more details are in the private review to the authors). The Results section is also quite long and bogs down at times, and I suggest that the authors reduce its length by 10 to 20%. In contrast, the Introduction is

sparse and lacks key aspects, for example, there should be mention of the study's main purpose and approaches, plus an introduction to the authors' choice of species and their known drug resistance properties, as well as the drug of choice (balofloxacin). Another notable weakness is that the authors evaluated Mg²⁺-induced phenotypic resistance only against two closely related species, and thus the generalizability of this mechanism of drug resistance is not known. The paper would be strengthened if the authors could demonstrate this type of phenotypic resistance in at least one more Gram-negative species and at least one Gram-positive species (antimicrobial susceptibility evaluations would suffice), each of which should be pathogenic to humans. Demonstrating magnesium-induced phenotypic drug resistance in the WHO Priority Bacterial Pathogens would be particularly important.

In general, the conclusions drawn by the authors are justified by the data, except for the interpretation of some experiments. Importantly, this paper has discovered new antimicrobial resistance mechanisms and has also pointed to potential new targets for antimicrobials.

<https://doi.org/10.7554/eLife.100427.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public Review):

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*In the manuscript entitled "Magnesium modulates phospholipid metabolism to promote bacterial phenotypic resistance to antibiotics", Li et al demonstrated the role of magnesium in promoting phenotypic resistance in *V. alginolyticus*. Using standard microbiological and metabolomic techniques, the authors have shown the significance of fatty acid biosynthesis pathway behind the resistance mechanism. This study is significant as it sheds light on the role of an exogenous factor in altering membrane composition, polarization, and fluidity which ultimately leads to antimicrobial resistance.*

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- (2) An adequate number of replicates were used for the experiments.*

Weaknesses:

- (1) The introduction section needs to be more informative and to the point.*
- (2) The weakest point of this paper is in the logistics through the results section. The way authors represented the figures and interpreted them in the results section (or the figure legends) does not match. The figures are difficult to interpret and are not at all self-explanatory.*
- (3) There are too many mislabeling of the figure panels in the main text which makes it difficult to find out which figures the authors are explaining. There should be more explanation on why and how they did the experiments and how the results were interpreted.*

- (1) We would like to extensive revise the introduction to make it more informative than the current version.*

- (2) We will check the description in the text and labeling in the figures to make it is logic.
- (3) We will add the explanation of the experiments to make it clear that why we perform the assays.

Reviewer #2 (Public Review):

Summary:

In this study, the authors aimed to identify if and how magnesium affects the ability of two particular bacteria species to resist the action of antibiotics. In my view, the authors succeeded in their goals and presented a compelling study that will have important implications for the antibiotic resistance research community. Since metals like magnesium are present in all lab media compositions and are present in the host, the data presented in this study certainly will inspire additional research by the community. These could include research into whether other types of metals also induce multi-drug resistance, whether this phenomenon can be observed in other bacterial species, especially pathogenic species that cause clinical disease, and whether the underlying molecular determinants (i.e. enzymes) of metal-induced phenotypic resistance could be new antimicrobial drug targets themselves.

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We will add the explanation of the experiments to make it clear that why we perform the assays. And we will revise the introduction and shorten the length of the manuscript. Expanding the bacterial species is very good idea and we will perform such experiment.

<https://doi.org/10.7554/eLife.100427.1.sa0>